

THESIS

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Preface

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1 Introduction

1.1 General background

The Covid 19 pandemic showed structural challenges of national economies all over the world, specifically the fragility of neoliberal policies in times of crisis and the lack of industrial and economic resilience. Six years after the fact, our societies are now confronted with inevitable novel challenges and looming shocks. We are already witnessing the consequences of AI development as it paves the way for a new technological revolution that would render most local economies obsolete and cause massive unemployment in white collar sectors. Sadly, that doesn't seem to be the end of the shocks the world has seen recently as the war in Ukraine, Gaza, Iran, and most importantly the current, chaotic trade wars, seem to foster ever increasing uncertainties. Facing all this, policy makers are confronted with a simple choice; strategise and plan for more resilient local economies. In different streams of literature, resilience is directly related to diversification/variety, whether in portfolio management in finance, trade partnerships/linkages, industrial activities, or in terms of knowledge as well. Thus we can equivalently say that resilience is the capacity

to resist and/or adapt to external shocks by relying on exiting internal capabilities that evolve in the face of such shocks. This means that for an economy to survive uncertainties, it needs to evolve, change, and innovate its way to the other end of the bleak challenges it's confronted with. However, to evolve, change and innovate, a baseline of knowledge should be leveraged since the consensus is that variety is a buffer against external shocks and shields uncertainties. This chain of thoughts, takes us back to the conceptual basics of sustainable economic development and growth; the knowledge fabric is what facilitate any long term strategy, and has been shown in the literature as clear catalyst for societal prosperity and economic resilience. For this reason the study of knowledge is detrimental for policy making, and understanding how to increase its diversity is more relevant than ever before.

Knowledge is a set of information that covers one or many topics, and its characteristics are contingent on the different forms it can take or how it was created, generally speaking, academia and businesses are the main knowledge creators in any economy through research and patents. Essentially, knowledge can be codified (accessible by anyone through any medium), or tacit (personal information based on social connections, intuition, experience, etc, that's hard to share with others). The consensus in the literature is that the main driver of competitive advantage for firms is the tacit form of knowledge, which is also widely acknowledged that it's space dependent. However, the knowledge produced by firms can be reliably seen in patents, although they capture codified information, they also reveal tacit knowledge and its geographic footprint in space. This means that its detrimental to assess the knowledge in patenting activities (we refer to this knowledge as

technologies), and more so to focus on the local aspect of these activities i.e: sub-national regions.

This framing, however, is not at all new or novel. In fact, this is the entire aim of the literature of the geography of innovation; to study how innovation is created and diffused to different actors in different geographical contexts. Specifically, relatedness and economic complexity (REC) is one of the main streams of literature that focus on the relationships between activities and geographies. The conceptual and methodological framework that REC provides is widely used and adopted in academia and among policy practitioners and was one of the main contributors to the smart specialisation policy literature. The ideas embedded in this framework, to put it simply, rely on the premise of spatial dependence of tacit knowledge in local/regional economies/geographies and focus on simplifying these relationships using network science to model the relationships between knowledge and regions. Albeit these simplifications provide valuable scope for analysis and interpretation, the cost from the loss of granular information implies that there's much more conceptual, methodological, and empirical work needed. The reason for this is because the loss of information bias the empirical interpretation in the sense that we end up with a homogeneous implication with weak regards to the regional and national contexts as well as the technological characteristics. This work is motivated by this gap and aims to simply contextualise the study of knowledge diversification using the same granular information publicly available and commonly used in the REC literature. The idea is simple, account for endogenous and exogenous contexts using granular data and understand the contexts and contingencies that drives regional diversification.

1.2 Problem statement

The main problem that this work aims to solve is directly embedded in the methodological and empirical framework of REC. REC models diversification through network aggregation based on co-location and co-occurrence patterns. Using these patterns, different aggregations are used to quantify relatedness (the frequency of observing a pair of activities in the same region), and relatedness density (how much of the activities frequently observed together a region has). However, these measures are often interpreted not as aggregations of frequent observations but rather as relationship models. Empirically, diversification is studied using these constructs as predictors of entry, that is a region's entry to a new specialisation in a new technology, often considered as a binary outcome. In here we briefly outline the big picture in the REC methodological and empirical framework, its conceptual issues, its empirical consequences, and highlight the research gap.

1.2.1 Relatedness, relatedness density, and diversification

Relatedness and relatedness density are essentially measures of proximity. In a sense they describe how close two technologies are close to each other, or how close a given technology is to a region given its portfolio of technologies. To further decompose the problem here, we will first establish the methodological constructs for proximity measures. For relatedness that's co-occurrence, and for relatedness density that's the linear aggregation of relatedness.

First, co-occurrence is essentially the frequency of observing two activities together. In

the REC literature, this frequency describes the strength of the relationship. Activities frequently observed together are more related than the pairs rarely observed together.

Second, linear aggregation of relatedness essentially measures the percentage of co-located technologies in a region that are related to a reference technology. Thus, we can think of relatedness density as the link between related technologies and co-located technologies. The idea in the REC literature states that relative to a given technology, the more related technologies a region has, the more likely that it can develop that technology.

These two constructs are used together to predict the probability that a region will enter a new technology. The REC literature shows that relatedness density is consistently associated with higher probabilities in almost all studies. These results among others were one of the major latent contributors to smart specialisation strategy (S3) policy. Thus the consensus in the literature was clear: In order to diversify into new technologies with the highest likelihood of success, regions must prioritise investment in related technologies.

1.2.2 Empirical consequences

The idea of resilience is not a main focus for the REC literature nor it is ours. However, falling back to this concept allows us to further assess the empirical consequences of the mainstream interpretation of relatedness and relatedness density. The idea is that in order to be resilient to external shocks and subsequent uncertainties diversification is key. But what kind of diversification is required and feasible and how to achieve it is the focus here. The REC literature tells us that the most likely successful diversification strategy

is the one that targets related capacity in the regions, often referred to as related variety. However, generalising this recommendation is not that straight forward. Aggregate regional capacities and their national and broader geographic contexts differ significantly. The initial landscape of the regional technological portfolio is detrimental here because this strategy could favour regions with already diverse portfolio but it's questionable that regions with limited portfolios would equally benefit. This is aligned with the concept of path dependency, related variety without context enforces that dependency and locks regions within their limited capacities. This brings us back to the core focus of in this work; context is key. However, context on its own here might not be enough since path dependency of related variety is a direct result of how relatedness and relatedness density is calculated and interpreted. The constructs that enable these measures (co-occurrence and linear aggregation) are the core problem that we're highlighting here. The reason behind this specific focus relied on the implicit assumptions embedded in these methodological constructs.

Co-occurrence assumes that technologies frequently observed together are likely related. Although this is within the boundaries of common sense, it's highly unlikely that's actually the case. A frequency measure is only informative when we have more observations than items—that is, more geographies than technologies. Almost always, we will have more technologies than geographies, this means that relatedness is at best noisy. Additionally, relatedness as interpreted in the literature quantifies the relationship between pairs of technologies. However, as it stands, there's no differentiation in the direction of that relationship thus, assuming that the relationship between two technologies is sym-

metrical. Albeit this assumption in itself is not problematic, it exacerbates the linearity issue when we measure relatedness density as we lose information in co-occurrence, symmetry, and linear aggregations. This takes us to the final issue we would like to highlight; relatedness density. Simply put, relatedness density measures the sum of technologies related to a reference technology present in a given region. The implicit assumption in here is that technologies are linked through linear combinations, and those combinations predict the likelihood of successful diversification. However, relatedness density is often interpreted as a value that quantifies the existing requirements a region has relative to a technology, whereas the sum of existing related technologies do not inform us on the actual requirements.

In summary, relatedness and relatedness density measures suffer from diverse methodological issues embedded in the implicit assumptions in their core constructs. Co-occurrence and linear aggregation of observed frequency are misinterpreted, accrues information loss, and poorly handles the granular data often used. This means that the empirical and methodological work ahead must account for these issues to further contextualise the study of diversification strategies.

1.2.3 Research problem

In the light of all the mentioned in this section, we fall back again on the core idea that we started this text with; How can we contextualise diversification strategies? The answer to this question is multi-layered and complex. In this section we started by outlining the importance of diversification strategies for regions into new technologies via patenting

activities. We explained that the REC literature provides interesting methodological and conceptual framework of analysis and showed that despite their usefulness they suffer from structural issues that limit the advantages of the used granular data, thus limit the incorporation of a broader context empirically. Essentially, the research problem we focus on here, is both methodological and empirical in nature. We highlighted the structural methodological issues as the research gap which will be the focus of our methodological and empirical contribution.

To address the methodological gaps in REC, we propose a multilayered contextual framework that distinguishes between different contextual factors. Relatedness measures alone conflate these dimensions. By decomposing context into endogenous technology-region fit and exogenous environmental conditions, we can identify which combinations enable successful diversification across different regional contexts, following this design:

Endogenous factors (within-region characteristics):

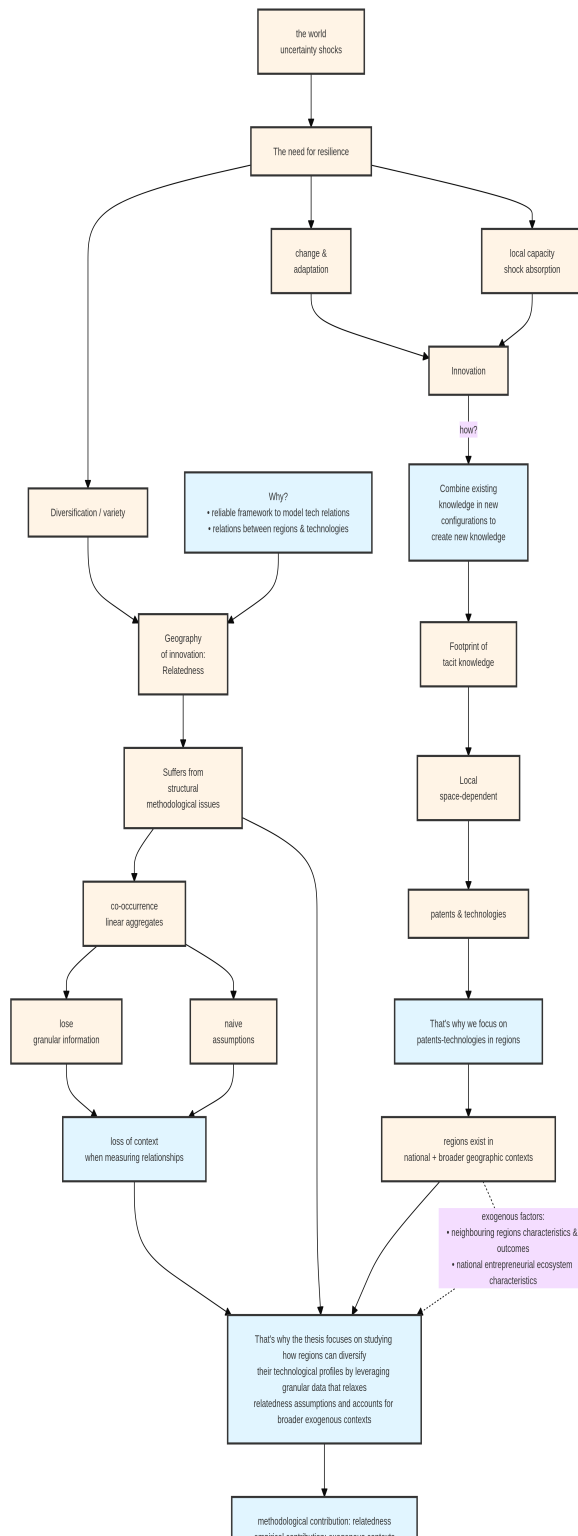
- Technology-specific attributes (network position, centrality, embeddedness)
- Regional knowledge infrastructure (coherence between technology relationships and regional specialization patterns)
- Regional technological volume (number of patents in a specific technology)

Exogenous factors (beyond-region influences):

- National entrepreneurial ecosystem characteristics (institutions, financing, R&D culture)
- Spatial spillovers from neighbouring regions (outcome diffusion, knowledge flows)

- Geographic proximity to coherent knowledge clusters

The interaction between these layers is critical: endogenous capacity determines what a region can develop, while exogenous context shapes realisation possibilities. For instance, a technology may align with regional knowledge infrastructure (endogenous coherence), but without supportive national institutions or proximity to innovative neighbours (exogenous support), entry probability remains low.



1.3 Objective

The objective from this work is to extend the methodological framework of relatedness starting from the structural issues it suffers from. In order to show how our methodological contribution benefit the study of regional technological diversification we rely on an empirical study that shows how broader context can be included and how such context inform granular policy insights. The broader context we aim to include empirically relied on relaxing the implicit relatedness assumptions, explicitly include the endogenous characteristics of the technologies and regions contingent on the regional knowledge infrastructure while simultaneously account for the characteristics of the national ecosystems. With such an approach we end up accounting for more contextual layers than the mainstream approach.

1.4 Research questions and hypothesis

RQ1: How to further contextualise diversification strategy based on the relatedness framework?

- H1a: Asymmetric relatedness measures better predict technology entry than symmetric measures.

RQ2: Is successful diversification contingent on regional knowledge infrastructure?

- H2a: Technology-specific characteristics impact on diversification is contingent on the regional knowledge coherence.
- H2b: Technology-specific characteristics impact on diversification is contingent on the regional knowledge stock.

RQ3: Does the national ecosystem influence diversification?

- H3a: National Entrepreneurial Ecosystem characteristics positively affects regional entry into new technologies.

RQ4: What role does space have? Do neighbouring regions influence diversification?

- H4a: Spatial spillovers of outcome from neighbouring regions positively affect technology entry.
- H4b: Neighbouring regions' technological characteristics influence focal region diversification outcomes.
- H4c: Geographic proximity to regions with coherent knowledge infrastructure increases entry probability.

1.5 Structure

We structure this work around our methodological and empirical contributions. Given the complex nature of the problems we aim to solve, we will first start with more literature context that underlines geography of innovation, and more importantly relatedness

and economic complexity literatures. We then outline our approach in the methodology chapter where we discuss in more details measures of relatedness and proximity and provide alternative conceptualisation to modeling the relationships between technologies and regions. In the same chapter we also detail other elements that will be relevant to the empirical part specifically knowledge coherence. In the empirical chapter we outline the research design and the modeling procedures that implements our methodological contribution. The results chapter outline the results of our work and its empirical consequences and interpretation, which we discuss in more detail in the discussion and conclusion of this work in which we also outline different further extensions and future research directions.

2 Literature review

2.1 Regional diversification as an outcome to resilience:

Overview from an Evolutionary perspective

In the previous chapter, we framed this work’s core idea around the need to study technological diversification patterns of regions within a broader context. To justify this framing we grounded this core idea on the ever increasing importance of resilience facing an increasingly uncertain global and geopolitical world order (Xiao, Boschma, and Andersson 2018). Although resilience in itself is not a core focus of this work, we will use it as an adjacent conceptual paradigm to justify our core ideas conceptually, methodologically and empirically through the literature.

Within the literature of evolutionary economic geography, P.-A. Balland, Rigby, and Boschma (2015) summarize this turn toward resilience and highlight that, while interest has grown, “the resulting debate suggests [that economic geographers] are far from a common understanding”. This is to say that resilience as a concept is attracting more

attention, however, there’s still no consensus on a generalised ontological framework that drives further research in this area. The different studies that addressed this concept (eg: Christopherson, Michie, and Tyler (2010); Fingleton, Garretsen, and Martin (2012); Martin (2012); Pendall, Foster, and Cowell (2010); Pike, Dawley, and Tomaney (2010); Simmie and Martin (2010)) relatively converge in refuting equilibrium based approaches that consider resilience as an immediate reaction to shock, and calling for further focus on an evolutionary approach. The argument for this proposed agenda is related to robust results showing that precedent regional capabilities are recycled over time with different configuration driving different regional growth paths (Boschma, Balland, and Kogler 2015; Neffke, Henning, and Boschma 2011). However, P.-A. Balland, Rigby, and Boschma (2015) also criticises the literature in this context for ignoring key explanatory dimensions that are central to evolutionary accounts of regional change; technological relatedness, network structures, and institutions.

This agenda matters for the present study because our goal is not to address regional resilience, but to explain the mechanisms that shape diversification feasibility in a way that is consistent with an evolutionary view of regional adaptation. Some work has been done in this context notably Tóth et al. (2022) who address the relatedness and network structures gaps. The authors anchor their view on resilience by considering it an “adaptive ability of regions to reconfigure their economic structure” and that this ability could be observed in diversification patterns. This intuition was addressed and confirmed in Rocchetta et al. (2022) for technological diversification patterns and for industrial diversification patterns in Xiao, Boschma, and Andersson (2018).

Against this background, we adopt an evolutionary framing that treats diversification as a constrained, path-dependent process of recombination. Empirically, this motivates a more contextualized assessment of regional capabilities and a decomposition of diversification drivers along three dimensions that the literature highlights as consequential: relatedness, network structure, and higher-level context (including national entrepreneurial ecosystems and spatial interdependence).

2.2 Geography of innovation: why place matters?

2.3 REC: what it achieves → what it assumes → where it fails

2.3.1 ML approach: how to fix the methodological problems

2.3.2 Coherence tradition: what's missing conceptually

2.4 Contextual layers: endogenous (FITS, coherence) + exogenous (ecosystem, space)

2.5 Our framework: positioned against the literature

3 The interconnectedness between regions and technologies

3.1 Methodological Motivation

Traditional relatedness frameworks model diversification through symmetric co-occurrence matrices and linear aggregation of relatedness density (Hausmann and Hidalgo 2011). However, as established in our problem statement, these constructs suffer from three structural limitations: (1) symmetry assumptions that obscure directional dependencies between technologies, (2) linear aggregation that loses granular information about technology-specific requirements, and (3) noise when technologies outnumber regions.

We address these limitations by replacing traditional measures with a machine learning training strategy based on the Random Forest algorithm (RF). Specifically, we use RF to generate: (1) **Feature Importance Technology Space (FITS)** following Fessina et al. (2024), analogous to the mainstream Technology Space. FITS captures directional, hierarchical technology relationships, replacing symmetric relatedness measures; and (2)

predicted probabilities (technological potential) following Albora et al. (2023), analogous to relatedness density. The technological potential estimates region-specific feasibility of technology adoption given the region’s own technological portfolio, replacing linear relatedness density. This approach enables the contextualization of diversification strategies by accounting simultaneously for technology-specific characteristics, regional knowledge infrastructure, and the broader non-linear interconnectedness between regions and technologies that traditional measures cannot adequately capture.

Before detailing our methodology and how we leverage the contributions in Fessina et al. (2024) and Albora et al. (2023), we will briefly situate this approach within the REC literature. Essentially, REC formalizes the empirical observation that shared input requirements (knowledge, resources, capabilities) determine diversification feasibility (Hidalgo et al. 2018; Hidalgo 2021). While this framework have proven valuable for policy (Zaccaria et al. 2018; E and A 2021), deriving granular, context-specific implications remains challenging (Hidalgo 2023; Li and Neffke 2024). Recent work on unrelated diversification (Flávio L. Pinheiro et al. 2022; Boschma et al. 2023), geographic inequalities (Flavio L. Pinheiro et al. 2025; Hartmann et al. 2017), and emerging technologies (Lee et al. 2018; Fessina et al. 2024) highlights the need for methodologies that capture contextual nuances beyond path dependency identification which we will elaborate on in subsequent sections.

Our approach is a response to the nuances expressed in the literature. But in order to highlight adequately our work we must first elaborate on the mainstream approach conceptually, methodologically and its empirical implications which we briefly outlined in the introduction. From there we expand on the mainstream approach and construct ours

conceptually, mathematically and showcase our training strategy. To this end, we also provide a brief example on how the mainstream approach and the machine learning approach to modeling the technology space compare. For the sake of clarity and consistency we will establish our notation system and the main definition in this paragraph and use them in the subsequent sections and chapter in this dissertation unless we specify otherwise.

We consider the following sets:

- $\mathcal{T} = \{t_i\}_{i=1}^{N_T}$: set of technologies
- $\mathcal{R} = \{r_i\}_{i=1}^{N_R}$: set of regions
- $\mathcal{Y} = \{y_i\}_{i=1}^{N_Y}$: set of years
- $\mathcal{C} = \{c_i\}_{i=1}^{N_C}$: set of technology categories
- $\mathcal{K} = \{\kappa_i\}_{i=1}^{N_K}$: set of countries

These sets are mirrored by the European Patent Office data (1978-2021) classified at the 4-digit IPC level, yielding 641 distinct technologies across 345 NUTS2 regions in 34 European countries. IPC classifications provide hierarchical structure: section (letter, we also refer to this as categories), class (two digits), subclass (letter). For example, F16H encompasses Section F (Mechanical engineering), Class 16 (engineering elements for mechanical power transmission), and Subclass H (gearing systems). We supplement patent data with Eurostat regional socio-economic indicators detailed in subsequent sections.

We quantify regional specialization using the Revealed Comparative Advantage (Balassa 1965). Although originally designed for trade data, this metric has been widely adopted

in innovation geography literature. Following the nomenclature in P.-A. Balland and Rigby (2017), we refer to it as Revealed Technological Advantage (RTA) in our patent data context. The RTA measures relative specialization, enabling simultaneous capture of expertise depth and portfolio diversity. Despite critiques regarding patent-based applications (P. Balland and Boschma 2019; Diodato et al. 2023), RTA aligns with our objective of capturing meaningful technology relationships through machine learning rather than raw co-occurrence. The RTA for region r in technology t during year y is defined as:

$$\text{RTA}_{r,t,y} = \frac{\frac{X_{r,t,y}}{\sum_{t'} X_{r,t',y}}}{\frac{\sum_{r'} X_{r',t,y}}{\sum_{r',t'} X_{r',t',y}}} = \frac{X_{r,t,y} \sum_{r',t'} X_{r',t',y}}{(\sum_{t'} X_{r,t',y}) (\sum_{r'} X_{r',t,y})}$$

Where, $X_{r,t,y}$: patent count for region r in technology t during year y

For each year y , we construct the RTA matrix $\mathbf{R}^{(y)}$ with entries $\text{RTA}_{r,t,y}$:

$$\mathbf{R}^{(y)} = [\text{RTA}_{r,t,y}]_{r=1,\dots,N_R}^{t=1,\dots,N_T}$$

These yearly matrices form the foundation for all subsequent modeling. We then binarize specialization for classification tasks:

$$z_{r,t,y} = \begin{cases} 1 & \text{if } \text{RTA}_{r,t,y} \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

where $z_{r,t,y} = 1$ indicates region r has comparative advantage (specialization) in technol-

ogy t at year y .

To further elaborate, let's consider $c_{t,t'}$, the count of patents containing both technologies t and t' ($t \neq t'$) at a given time period, such that $t, t' \in \mathcal{T}$, and $c_t, c_{t'}$ denote the total patent counts for each class individually.

In the following we will begin by outlining how the REC literature approach relatedness, how it's calculated and discuss in detail different technical contributions and their limitations. We will then present our approach and detail our methodological and conceptual contribution in response to these limitations.

3.2 The mainstream approach

3.2.1 Relatedness Through Co-occurrence

The REC literature operationalises relatedness through the observed frequency of co-occurrences. The intuition is appealingly simple: technologies that frequently appear together—whether in the same patent, the same firm, or the same region—likely share underlying knowledge requirements. If mastering one technology facilitates developing another, we should observe them co-located more often than chance would predict. This reasoning has proven influential, providing the empirical foundation for technology space visualisations and diversification recommendations across academic and policy settings.

Yet the inference from co-occurrence to capability-based relatedness rests on assumptions that warrant scrutiny. Co-occurrence is a frequency measure; it tells us how often two

technologies appear together, not why they do so or whether that co-presence reflects shared inputs, complementary applications, or statistical artefact. The distinction matters: a measure intended to guide capability-building strategies must capture genuine knowledge relationships rather than patterns that emerge from data structure alone.

Let $c_{t,t'}$ denote the count of patents containing both technologies t and t' (where $t \neq t'$) within a given time period, and let c_t and $c_{t'}$ denote the total patent counts for each technology individually. The simplest relatedness measure would be $c_{t,t'}$ itself, but raw co-occurrence counts are problematic: frequently patented technologies will naturally co-occur more often simply due to their prevalence. Various normalisation schemes address this issue.

Association Strength compares observed co-occurrence to expected co-occurrence under independence:

$$\phi_{t,t'}^{\text{assoc}} = \frac{c_{t,t'} \cdot T}{c_t \cdot c_{t'}}$$

where $T = \sum_{t \in \mathcal{T}} c_t$ is the total patent count. Values exceeding one indicate that the technologies appear together more frequently than would be expected if their occurrences were independent—they are positively associated. This measure properly corrects for size effects: a rare technology and a common technology can still exhibit strong association if they co-occur more than their marginal frequencies would predict.

The Probability Index refines this probabilistic approach by accounting for the fact that a patent cannot co-occur with itself:

$$\phi_{t,t'}^{\text{prob}} = \frac{c_{t,t'}}{\frac{1}{2} \left[\frac{c_t}{T} \cdot \frac{c_{t'}}{T-c_t} + \frac{c_{t'}}{T} \cdot \frac{c_t}{T-c_{t'}} \right] \cdot T}$$

This correction, while subtle, provides more accurate estimates for patent co-occurrence analysis and is implemented in standard software packages used in the field.

Cosine Similarity offers a set-theoretic alternative:

$$\phi_{t,t'}^{\text{cosine}} = \frac{c_{t,t'}}{\sqrt{c_t \cdot c_{t'}}}$$

This measures relative overlap between technology profiles. While intuitive, it systematically favours frequent technologies over rare ones and does not properly correct for size effects, making comparisons across technologies with different prevalences problematic.

Minimum Conditional Probability takes a different approach by examining regional co-location rather than patent-level co-occurrence:

$$\phi_{t,t'}^{\text{min}} = \min \{P(\text{RTA}_{r,t}|\text{RTA}_{r,t'}), P(\text{RTA}_{r,t'}|\text{RTA}_{r,t})\}$$

This computes the minimum of the conditional probabilities: given that a region specialises in one technology, how likely is it to specialise in the other? The minimum operator ensures symmetry while avoiding inflation from imbalanced conditional relationships.

A clarification is warranted here regarding two distinct but often conflated concepts. Co-occurrence typically refers to technologies appearing together within the same patent or

firm, while co-location refers to technologies appearing together within the same region. Both capture related but different phenomena: co-occurrence within patents reveals direct technical complementarities, while co-location within regions reveals shared capability requirements that may operate at greater conceptual distance. The literature sometimes uses these terms interchangeably, but the distinction matters for interpreting what relatedness measures actually capture.

3.2.2 Structural Limitations of Co-occurrence-Based Relatedness

Despite the variety of normalisation approaches, co-occurrence-based relatedness measures share structural limitations that affect their reliability and interpretation.

Dimensional mismatch produces noisy estimates. The reliability of frequency-based measures depends on having sufficient observations to estimate co-occurrence patterns accurately. In typical applications, the number of technologies substantially exceeds the number of locations—estimating a matrix of several thousand technology pairs from fewer than two hundred country observations, for instance, produces correlation structures that are largely random (Tacchella et al. 2023). The number of observations required for reliable estimation of such matrices easily reaches tens of thousands (Bun, Bouchaud, and Potters 2017), far exceeding available data. This problem is not resolved by moving to subnational levels, where harmonised data is often unavailable and the fundamental dimensional mismatch persists.

Nested structure dominates the relatedness signal. Location-activity networks typically exhibit strong nested structure: diversified regions produce almost everything while

less diversified regions produce only ubiquitous activities (Mariani et al. 2019). This triangular pattern means that when counting co-occurrences, one primarily detects the diversification hierarchy rather than genuine capability-based relationships. The relatedness signal becomes second-order relative to the structural pattern that generates nestedness (Tacchella et al. 2023). This explains why even randomised relatedness matrices produce predictions comparable to co-occurrence-based measures—they still capture the fact that diversified regions are likely to add activities regardless of which specific activities are involved (Bustos et al. 2012).

Spurious associations are rarely filtered. Studies employing co-occurrence measures often fail to test the statistical significance of the relationships they identify (Saracco et al. 2015, 2017). Without appropriate null models, spurious co-occurrences due to sampling variability are mistaken for genuine relatedness. When such tests are applied to trade data, the results are sobering: the null hypothesis of random diversification—no path dependence—is rejected for only approximately half of countries examined, with most high-income and large economies exhibiting diversification patterns that defy path dependence entirely (Coniglio et al. 2021). Whether these findings extend to patent-based technology data remains an open question, but they raise concerns about the generalisability of path-dependent diversification as a universal empirical regularity.

Symmetry obscures directional dependencies. All normalisation schemes above produce symmetric measures: $\phi_{t,t'} = \phi_{t',t}$. This symmetry assumes that the relationship between any two technologies operates bidirectionally with equal strength. Yet technological development often exhibits clear hierarchies (Boschma 2017). Expertise in semicon-

ductor fabrication may strongly predict future development of integrated circuit design, but the reverse need not hold—circuit designers may not subsequently develop fabrication capabilities. The capabilities required to transition from technology i to technology j may differ vastly from those required for the reverse transition. Symmetric measures cannot distinguish prerequisite relationships from derived relationships, enabling technologies from their applications, or identify the stepping-stone pathways that matter for strategic planning.

3.2.3 Relatedness Density Through Linear Aggregation

Relatedness density (Hidalgo et al. 2007) bridges the gap between pairwise technology relationships and region-specific diversification potential. Where co-occurrence measures describe technology-to-technology proximity, relatedness density aggregates these proximities relative to a region’s existing portfolio, producing a single value meant to capture how feasible a new technology is for a given region. The measure has become central to diversification analysis and policy guidance, serving as the primary predictor in entry models and the basis for strategic recommendations.

The aggregation, however, imposes strong assumptions about how capabilities combine. By summing pairwise relationships, relatedness density treats the regional portfolio as a linear accumulation of independent components. Whether this adequately represents how knowledge and capabilities actually interact in enabling new activities is an empirical question that the measure’s structure forecloses.

For region r and technology t , relatedness density aggregates the relatedness between technology t and all technologies in which the region currently specialises:

$$\text{relatedness density : } \Phi_{r,t} = \frac{\sum_{t' \in \mathcal{T}_r, t' \neq t} \phi_{t,t'}}{\sum_{t' \neq t} \phi_{t,t'}} \times 100$$

where $\mathcal{T}_r$ denotes the set of technologies in which region r specialises (those with $\text{RTA}_{r,t} \geq 1$).

The interpretation is intuitive: the numerator sums relatedness values to technologies the region already has, while the denominator normalises by total relatedness to all technologies. High relatedness density indicates that technology t is “close” to the region’s existing capabilities—many of its related technologies are already present in the regional portfolio. Low relatedness density suggests that t requires capabilities the region currently lacks.

This measure has proven remarkably predictive: across numerous studies using different data, time periods, and geographic contexts, higher relatedness density consistently associates with higher probability of future entry. Regions are more likely to develop specialisations in technologies that relate to what they already do well.

3.2.4 Structural Limitations of Linear Aggregation

The predictive success of relatedness density is not in question. The limitations concern what it cannot reveal and how its structure constrains interpretation.

Linear aggregation obscures combinatorial information. Relatedness density sums

pairwise relationships without regard to their interactions. This implicitly assumes that having two related technologies is simply twice as beneficial as having one—capabilities combine additively. Yet capabilities may interact in ways that linear summation cannot capture. Certain combinations may be more than the sum of their parts, creating complementarities where the joint presence of technologies A and B enables technology C in ways that neither alone provides. Other combinations may be redundant, contributing less than their individual relatedness values would suggest. By reducing many-body relationships to independent two-body terms, linear aggregation discards precisely the combinatorial information that may distinguish feasible diversification paths from infeasible ones (Tacchella et al. 2023). Empirically, prediction models that capture many-product interactions substantially outperform density-based predictions, suggesting that linear aggregation loses significant information (Albora et al. 2023).

Noise compounds through aggregation. The dimensional mismatch and spurious associations affecting pairwise co-occurrence estimates do not disappear when aggregated into relatedness density—they compound. Summing noisy pairwise values produces a noisy aggregate. The normalisation by total relatedness provides some correction but cannot recover signal that was lost at the pairwise level. For regions with portfolios concentrated in rare technologies, where pairwise estimates are least reliable, relatedness density inherits and potentially amplifies these estimation problems.

Frequency of entry conflates observation with feasibility. The empirical regularity that higher relatedness density associates with higher entry probability is often interpreted as evidence that related diversification is more feasible and therefore should be priori-

tised. But this inference conflates observed frequency with success rate as emphasised in C. Pinheiro (2025). We observe which entries occurred, not which were attempted; the denominator—attempts—remains unobserved. Related diversification may appear more common because it is attempted more frequently, perhaps precisely because existing frameworks recommend it, rather than because it succeeds more often when attempted. Coniglio et al. (2021) addresses precisely this issue and shows that related diversification attempts succeeded 80% in Germany in contrast to 40% in the United States and France. Moreover, the observed product and technology spaces may themselves reflect past policies promoting related diversification (Andreoni and Chang 2019), creating circularity when used to justify future policy.

3.2.5 Implications for Contextualisation

The limitations identified above are not independent—they compound. Noisy co-occurrence estimates feed into symmetric matrices that are then linearly aggregated, producing context-free measures whose predictive success may reflect structural properties of the data (nestedness, diversification hierarchies) as much as genuine capability-based relationships. The resulting framework provides a supply-side instrument: it identifies what regions might produce given their existing portfolios, while remaining silent on demand dynamics, competitive pressures, and the institutional conditions that shape whether capability-based proximity translates into actual entry (C. Pinheiro 2025; Andreoni and Chang 2019).

This supply-side orientation has consequences. When applied to policy, relatedness-based

recommendations may systematically favour already-diversified regions whose dense portfolios generate high relatedness density across many targets, while offering limited guidance to less-diversified regions whose sparse portfolios produce low density values regardless of strategic potential. The concern that such recommendations reinforce rather than ameliorate path dependency—locking peripheral regions into constrained trajectories while core regions accumulate further advantages—has been explicitly raised in recent literature examining regional inequality and the distributional consequences of relatedness-based strategies (Flavio L. Pinheiro et al. 2025; Mealy and Coyle 2022; Hidalgo 2023).

The issue is not that relatedness is empirically wrong—the predictive regularity is robust—but that the measures as currently constructed cannot inform us about the conditions under which relatedness matters more or less, about which technologies serve as stepping stones toward others, or about how regional knowledge infrastructure, national ecosystems, and spatial factors moderate diversification possibilities. Addressing these limitations requires reconstructing the core methodological constructs in ways that relax the problematic assumptions while preserving the fundamental insight that prior capabilities shape future possibilities. This reconstruction motivates the machine learning approach developed in the following sections.

3.3 An Alternative Approach: Machine Learning for Contextualised Relatedness

The limitations outlined above—dimensional mismatch producing noisy correlations, symmetric measures obscuring directional dependencies, and linear aggregation discarding interaction effects—are not merely technical inconveniences. They represent fundamental obstacles to using relatedness for prediction and policy. If co-occurrence methods perform no better than trivial baselines, as demonstrated in Tacchella et al. (2023), then their assessment of diversification feasibility offers insufficient guidance for strategic decisions. This motivates a shift from descriptive correlation to predictive modelling.

3.3.1 Machine Learning as Prediction Framework

The core reframing is methodological: rather than inferring relatedness from co-occurrence patterns and hoping this correlates with future outcomes, we directly model diversification as a prediction problem. This follows recent work positioning relatedness estimation within supervised learning frameworks (Tacchella et al. 2023; Albora et al. 2023; Fessina et al. 2024). The shift offers three advantages.

First, it provides a falsifiable evaluation standard. Co-occurrence measures proliferate without systematic comparison—proximity, association strength, minimum conditional probability, and numerous variants coexist with no principled basis for selection. Out-of-sample prediction performance offers an objective criterion: methods that better forecast actual diversification outcomes are, by definition, better capturing whatever underlying

relatedness structure matters for real transitions (Albora et al. 2023). This addresses the circularity where relatedness is inferred from outcomes and then used to explain those same outcomes.

Second, prediction-based approaches can capture the many-body correlations that density-based measures discard. As Tacchella et al. (2023) emphasise, describing diversification paths as sums of binary relatedness relationships is an oversimplification. A region’s probability of developing technology i depends not just on pairwise similarities but on the full configuration of its existing portfolio—which combinations are present, which are absent, and how these interact. Tree-based algorithms learn precisely these complex conditional patterns.

Third, the framework naturally accommodates the asymmetric, directed relationships that symmetric co-occurrence measures cannot represent. When we train separate models for each target technology, we obtain directional importance scores: technology t may strongly predict entry into technology i while i weakly predicts entry into t . This asymmetry reveals hierarchical structure—prerequisites, enabling technologies, and developmental sequences—that bidirectional similarity metrics obscure.

We apply this framework through two complementary constructs:

Technological Potential estimates $p_{r,t,y}$, the probability that region r develops specialisation in technology t by year y , given its existing portfolio. This replaces relatedness density. Rather than linearly summing symmetric co-occurrence frequencies, we predict region-specific feasibility using the full structure of past portfolios. The measure captures non-linear interactions and regional heterogeneity that additive approaches miss.

Feature Importance Technology Space (FITS) extracts directional dependencies $I_{t \rightarrow t'}$ between technologies from the trained models. This replaces symmetric technology networks. Instead of co-occurrence frequencies, we identify which technologies predict others’ future adoption—and crucially, which do not predict in the reverse direction. The resulting asymmetric network reveals hierarchical technological structure invisible to symmetric measures.

3.3.2 Why Random Forest

Among supervised learning approaches, tree-based ensemble methods—particularly Random Forest—offer properties well-suited to relatedness estimation. We briefly outline these before presenting the technical details.

High-dimensional sparse data. Regional technology portfolios are high-dimensional (hundreds of technology classes) and sparse (most regions specialise in few technologies). Tree-based methods handle this naturally through recursive partitioning: each split considers only a subset of features, and the hierarchical structure means irrelevant technologies simply do not appear in decision paths. Linear methods struggle with the collinearity and sparsity typical of specialisation data; neural networks require larger samples than regional patent data typically provide.

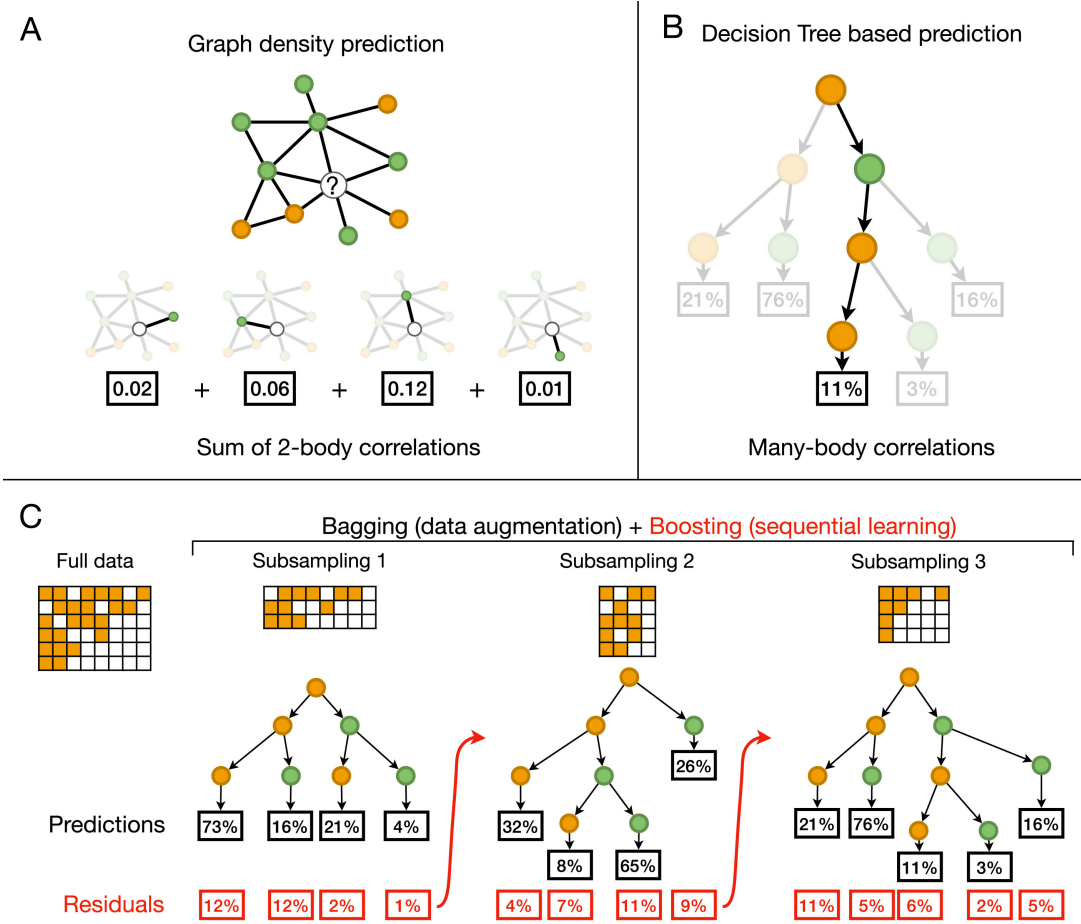
Non-linear interactions. The recursive splitting structure captures interactions without explicit specification. If technology t_1 matters for predicting t_3 only when t_2 is also present, this emerges naturally as a conditional split pattern. Density-based measures, by

construction, cannot represent such conditionality—they assume each related technology contributes independently.

Interpretable feature importance. Unlike black-box models, Random Forests produce feature importance scores that aggregate each predictor’s contribution across all trees. These scores become the basis for FITS construction, translating prediction performance into an interpretable technology network. This maintains the explanatory value of relatedness measures while grounding them in predictive validity.

Ensemble variance reduction. Bootstrap aggregation across many trees reduces the variance that would plague individual decision trees, particularly given the moderate sample sizes (regions $\times$ years) available in patent data. The ensemble structure provides stability without sacrificing the flexibility needed to capture complex capability interactions.

The choice of Random Forest over alternatives thus reflects both practical constraints (sample size, interpretability requirements) and the specific structure of relatedness estimation (sparse, high-dimensional, interaction-rich). We now present the algorithm formally before describing our application.



3.3.3 Random Forest Algorithm

Random Forest constructs an ensemble of B decision trees through bootstrap aggregating (bagging) with random feature subsampling. Each tree T_b is built on a bootstrap sample $\mathcal{D}_b^*$ drawn with replacement from the original dataset.

Each tree recursively partitions the feature space through binary splits. At node t , we randomly select m features (typically $m = \sqrt{p}$) and evaluate all possible splits within this subset. For feature j and threshold τ , the split creates two child nodes: $t_L = \{i : x_{ij} \leq \tau\}$ and $t_R = \{i : x_{ij} > \tau\}$.

Split quality is assessed via Gini impurity:

$$G(t) = 1 - \sum_{k=0}^1 p_k^2(t) = 2p_0(t)p_1(t)$$

where $p_k(t) = n_k(t)/n(t)$ represents the proportion of class k observations at node t . Gini impurity quantifies node heterogeneity: $G = 0$ indicates perfect purity (homogeneous class), while $G = 0.5$ indicates maximum impurity (equal class distribution). The optimal split maximises weighted impurity reduction:

$$\Delta G(j, \tau) = G(t) - \left[\frac{n(t_L)}{n(t)} G(t_L) + \frac{n(t_R)}{n(t)} G(t_R) \right]$$

Weighting by relative node size prevents trivial splits that isolate single observations into pure but uninformative leaves.

Recursive splitting continues until predefined stopping criteria: node purity ($G = 0$), minimum node size threshold, or maximum tree depth. Terminal nodes are assigned the majority class of their constituent observations.

For prediction, observation $\mathbf{x}$ traverses all B trees. Final classification aggregates individual tree predictions via majority voting:

$$\hat{y}(\mathbf{x}) = \text{mode}\{\hat{y}_1(\mathbf{x}), \dots, \hat{y}_B(\mathbf{x})\}$$

Class probabilities are estimated as the proportion of trees predicting each class:

$$\hat{P}(y = 1|\mathbf{x}) = B^{-1} \sum_{b=1}^B \mathbb{I}[\hat{y}_b(\mathbf{x}) = 1]$$

This probability represents empirical vote share across trees. Values near 1 indicate strong consensus for class 1 (high confidence), while values near 0.5 reflect uncertainty. Unlike parametric models, these are data-driven vote proportions rather than model-based probability estimates.

The algorithm’s effectiveness stems from variance reduction through decorrelated predictions. Bootstrap sampling and random feature selection reduce inter-tree correlation ρ , yielding ensemble variance:

$$\text{Var}(\bar{y}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

As B increases and ρ decreases, ensemble variance diminishes while maintaining the low bias of flexible tree models.

Feature importance quantifies each predictor’s contribution by aggregating Gini impurity reductions:

$$I(j) = \frac{1}{B} \sum_{b=1}^B \sum_{t \in T_b: v(t)=j} \Delta G(t)$$

where the sum runs over all nodes using feature j for splitting. Higher values indicate features consistently creating purer partitions. Feature importance measures predictive

association rather than causal effect, and exhibits bias toward high-cardinality features and correlated predictor sets. It ranks predictive relevance but requires caution in causal interpretation.

3.3.4 Training Procedure

We follow the methodology of Fessina et al. (2024) and Albora et al. (2023), originally developed for trade data. The key innovation lies in training separate models for each technology rather than constructing a single global model. For every technology $i \in \mathcal{T}$, we train a binary classification model where:

- **Outcome:** $z_{r,i,y}$ indicates whether region r specialises in technology i at year y ($\text{RTA} \geq 1$)
- **Features:** $\{\text{RTA}_{r,t,y-\delta} : t \in \mathcal{T}, t \neq i\}$ comprises RTA values for all other technologies δ years prior

This technology-specific approach enables each model to learn unique dependency patterns. The lag $\delta = 4$ years balances predictive horizon with data availability, following conventions in the literature on capability accumulation (Andreoni and Chang 2019).

We predict specialisation for target years $y_t \in \{2008, \dots, 2018\}$ using data from 1978 onward. For each target year y_t and technology i :

Training set:

$$X_{\text{train}} = \{\text{RTA}_{r,t,y} \mid y \in [1978, y_t - 2\delta], t \neq i\}$$

$$Y_{\text{train}} = \{z_{r,i,y} \mid y \in [1978 + \delta, y_t - \delta]\}$$

Test set:

$$X_{\text{test}} = \{\text{RTA}_{r,t,y_t-\delta} \mid t \neq i\}$$

$$Y_{\text{test}} = \{z_{r,i,y_t}\}$$

This temporal structure ensures strict separation: models predict future specialisation using only information available at the prediction date. The expanding training window incorporates historical diversification patterns while the fixed lag prevents information leakage.

The procedure yields 7,051 models ($641 \text{ technologies} \times 11 \text{ years}$). Given computational constraints, we performed cross-validation on a stratified sample of four technologies (G06G, B67B, D02J, C08J) spanning different IPC sections, applying the modal optimal parameters across all models: `mtry = 139`, `trees = 100`, `min_n = 38`. Training was conducted in R using the Ranger package (Wright and Ziegler 2017), orchestrated via the targets pipeline (Landau 2021).

3.4 Technological Potential

Each model produces probabilities $p_{r,i,y} = P(z_{r,i,y} = 1 \mid \text{RTA}_{r,\cdot,y-\delta})$ representing the likelihood that region r develops specialisation in technology i by year y , conditional on

its prior portfolio. These probabilities constitute **Technological Potential**—a forward-looking, region-specific measure of diversification feasibility.

The shift from correlation to prediction reframes what the measure captures. Density asks: “How similar is technology i to what region r already has?” Potential asks: “Given everything we know about how regions diversify, how likely is region r to develop technology i ?” The latter question—the one relevant for policy—requires a predictive approach.

High potential ($p_{r,t,y} \approx 1$) indicates a region’s existing portfolio strongly predicts future specialisation—the region likely possesses necessary complementary capabilities. Low potential ($p_{r,t,y} \approx 0$) suggests capability gaps that make diversification unlikely under current conditions. Intermediate values reflect genuine uncertainty, often indicating that diversification is possible but contingent on factors beyond current portfolio composition.

Crucially, potential varies across regions for the same technology. Two regions with similar aggregate relatedness density may have quite different potential scores because their specific portfolio configurations interact differently with the target technology’s requirements. This variation enables analysis of how regional context—knowledge infrastructure, institutional environment, spatial position—moderates diversification feasibility.

3.5 Feature Importance Technology Space (FITS)

Traditional technology networks rely on patent citations or co-occurrence patterns, each with limitations. Citations suffer from examiner additions that may not reflect actual

knowledge flows, aggregation from patent-level to technology-level that obscures directionality, and backward-looking orientation that captures historical influence rather than predictive relationships (Fessina et al. 2024). Co-occurrence networks inherit the problems outlined in our critique: symmetry, noise, and linear assumptions.

FITS addresses these by constructing an asymmetric, predictive network from the feature importance scores of our Technological Potential models. Rather than inferring relationships from co-occurrence, FITS extracts directional dependencies revealed by which technologies predict others' future adoption.

Recall that for each technology i , we trained a Random Forest model predicting $z_{r,i,y}$ using $\{\text{RTA}_{r,t,y-\delta} : t \neq i\}$ as features. The feature importance $I_i(t)$ quantifies how much technology t contributes to predicting future specialisation in technology i across all regions and time periods.

We formalise FITS as a directed, weighted network $G = (V, E, W)$ where:

- **Nodes** $V = \mathcal{T}$: the set of 641 technologies
- **Edges** E : directed connections $(t \rightarrow i)$ for all $t, i \in \mathcal{T}, t \neq i$
- **Weights** $W_{t \rightarrow i} = I_i(t)$: feature importance of technology t in the model predicting technology i

We normalise weights within each target technology:

$$W_{t \rightarrow i} = \frac{I_i(t)}{\sum_{t' \neq i} I_i(t')}$$

ensuring incoming edge weights sum to 1 for each technology. This normalisation facilitates comparison across technologies with different overall predictability.

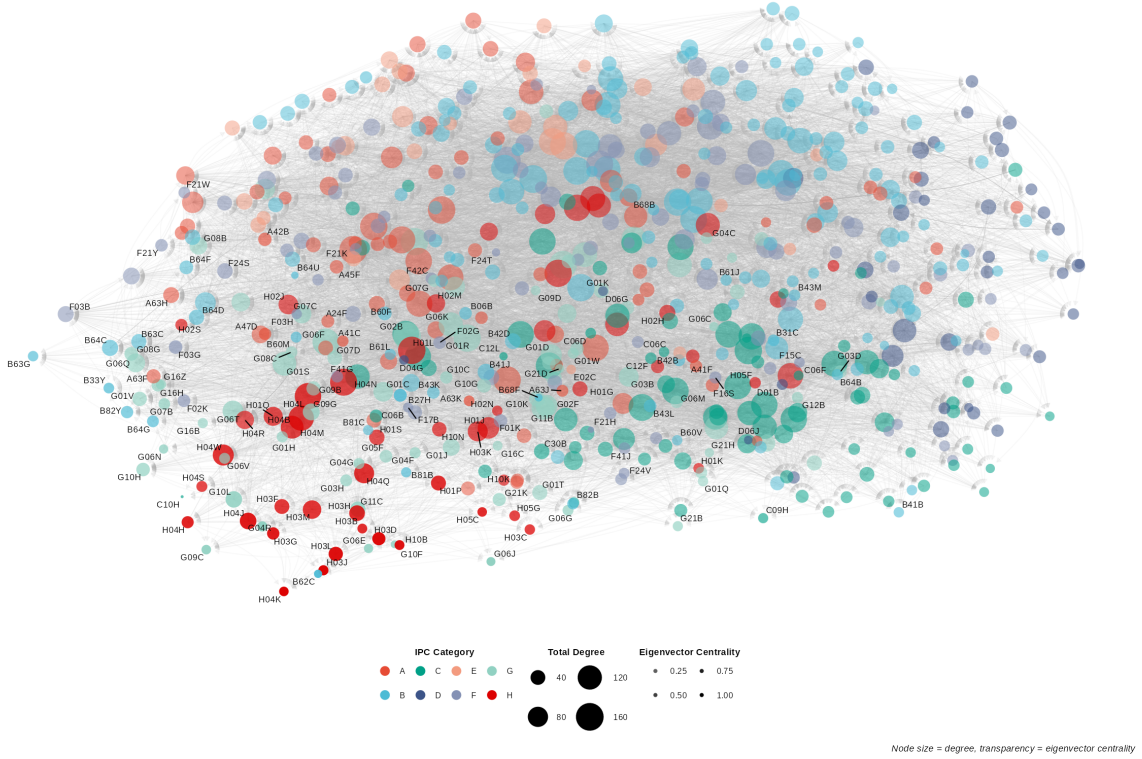


Figure 3.1: Technology Network (based on 2018 patent data)

The defining property of FITS is asymmetry: $W_{t \rightarrow i} \neq W_{i \rightarrow t}$ in general. This asymmetry captures hierarchical technological dependencies invisible to symmetric measures:

- $W_{t \rightarrow i} \gg W_{i \rightarrow t}$: Technology t is a prerequisite or enabler for i . Expertise in t predicts future development of i , but not vice versa. This pattern identifies foundational technologies and developmental sequences.
- $W_{t \rightarrow i} \approx W_{i \rightarrow t}$: Technologies are complementary peers with mutual predictive relationships, suggesting shared capability requirements or co-evolution.

- $W_{t \rightarrow i} \ll W_{i \rightarrow t}$: Technology i enables t , reversing the hierarchical relationship.

This directional structure reveals technological trajectories: regions can identify which current capabilities open paths toward desired future technologies, and which apparent similarities mask asymmetric dependencies.

The contrast between co-occurrence relatedness and FITS is sharpest when examining specific technologies. Consider H01L (semiconductors) and D06Q (textile decoration).

For semiconductors, co-occurrence relatedness identifies technologies that frequently appear together in regional portfolios. Some connections are sensible: semiconductor memories (H10B), packaging technologies (B68B). But the measure also surfaces implausible links—textile decoration (D06Q), woodworking (B27H), fusion reactors (G21B)—that appear not because of genuine technological affinity but because they happen to co-occur in diverse portfolios. This is the noise problem: frequency-based aggregation treats all co-presence equally, regardless of whether it reflects shared capabilities or statistical coincidence.

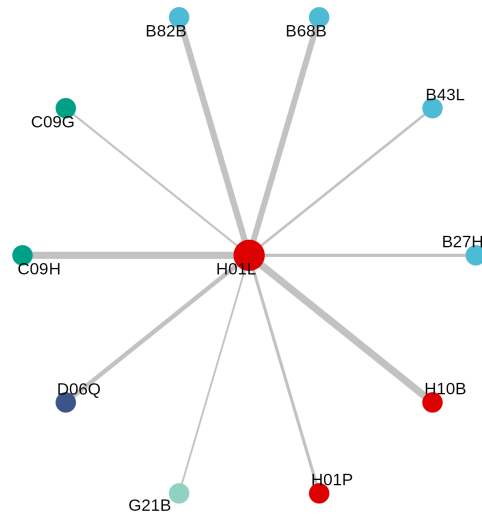
FITS reveals a different structure. Rather than asking which technologies appear alongside semiconductors, it asks which technologies predict future semiconductor development—and which technologies semiconductors predict. The asymmetry is striking: strong edges flow FROM application domains (digital computing G06F, telecommunications H04L, control systems G05F) TO semiconductors, indicating these fields depend on semiconductor capabilities. Weaker reverse edges show that semiconductor expertise does not particularly predict entry into computing or telecommunications. H01L emerges as an enabling technology whose importance lies in what it makes possible rather than what leads to it.

The contrast sharpens further with D06Q (textile decoration), a sparse technology where co-occurrence produces near-total noise: elevators (B64B), timing devices (G04D), chemical production (C13B) appear simply because D06Q is rare and co-occurs randomly with whatever else rare-technology regions happen to have. FITS filters this effectively, identifying genuine textile domain relationships—sewing (D05B), textile treatment (D06M), wall coverings (D06N)—that share actual functional connections.

This example illustrates how FITS captures the functional architecture of innovation networks that co-occurrence frequencies obscure. The method’s ability to distinguish signal from statistical artifact, combined with its directional structure revealing technological hierarchies, provides the foundation for contextualised analysis of regional diversification patterns.

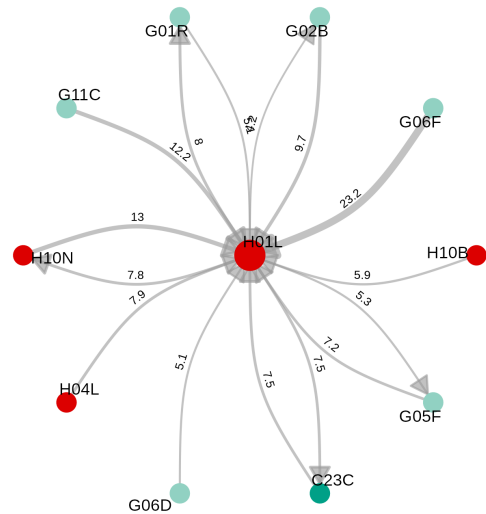
Relatedness Network

Co-occurrence based relationships to H01L



FITS Network

Bidirectional predictive relationships with H01L



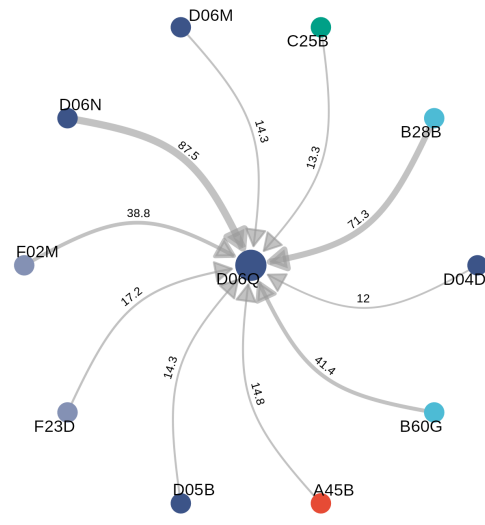
Relatedness Network

Co-occurrence based relationships to D06Q



FITS Network

Bidirectional predictive relationships with D06Q



Technological Potential provides region-specific feasibility estimates grounded in predictive performance rather than descriptive correlation. FITS reveals the directional, hierarchical structure of technological dependencies that symmetric measures cannot represent. Together, these constructs enable the contextualised analysis our research questions require: examining how regional knowledge infrastructure, national innovation systems, and spatial factors moderate the relationship between technological characteristics and diversification outcomes.

3.6 Coherence: Bridging Technology Networks and Regional Portfolios

Till now we established few points. First, FITS identifies directed technology-to-technology dependencies. And second, Regional Technological Potential quantifies region-specific feasibility of future specialisation. To bridge these two layers of information we introduce here the Coherence measure. Coherence here links these two levels by measuring whether a focal technology’s network profile matches the aggregate network profile of the region’s existing portfolio.

This logic follows the core idea in the coherent diversification tradition as established in (teece1994understanding?) and later in (nesta2005coherence?). The argument here is that performance is shaped not only by “how much” an actor does (diversification/variety), but also by how the activities in its portfolio hang together. Regional coherence in this sense is a structured property of the knowledge base, or as we referred to in the introduction as knowledge infrastructure. This logic aligns with recombinant views of innovation in which knowledge advances through combinations among related elements rather than as a homogeneous “stock” input (quatraro2010?). At the firm level, the same conceptual point is made sharply: two firms with the same number of technological fields can differ substantially depending on whether those fields form a coherent block or scattered knowledge infrastructure, coherence captures that internal structure (pugliese2019?).

3.6.1 Why we introduce coherence here (and why at IPC-section scale)

A recurring open issue in the literature is that the meaning and effect of coherence depend on scale and context. Recent studies shows that coherence can correlate positively with growth at metropolitan scales but turn negative at larger regional or national scales, consistent with the idea that at broader scales coherence may signal path dependence and lock-in rather than productive focus ([straccamore2025?](#)). This core disagreement with other relevant studies ([quatraro2010?](#); [pugliese2019?](#)) raises another challenge as it creates a practical problem for regional diversification analysis: if coherence behaves differently across scales, we need a measure that keeps coherence informative at the regional level without letting it collapse into a purely macro “lock-in” indicator.

Our solution is to bridge the scale disagreement by measuring coherence for regions at a smaller internal scale: instead of computing coherence as a single region-wide property, we compute coherence within broad technology domains (IPC sections, i.e. 1st digit code). The intuition is simple: regional portfolios can be coherent in some domains and not others, and those within-domain alignments are precisely what should matter for whether a region can realistically leverage a technology’s prerequisites and downstream connections in FITS. This is also consistent with innovation geography’s emphasis on “cognitive proximity” and the idea that relatedness works through structured overlaps in knowledge bases rather than mere activity counts (Neffke, Henning, and Boschma 2011; Boschma 2015).

To illustrate this further, let’s consider two regions, both lacking specialization in technol-

ogy t , both having similar potential $p_{r,t,y}$. However, Region A specializes in technologies that are strong predictors of t (strong weighted links), while Region B specializes in technologies unrelated to t in the network (weak weighted links). Coherence captures this difference: if region A has high coherence with t then its portfolio aligns with t 's network prerequisites. Similarly region B's portfolio is misaligned with t thus it would have low coherence.

This is directly aligned with the coherent diversification logic: what matters is not only how many technologies a region have, but whether its technology portfolio forms a coherent capability block at industry level around the target technology (**pugliese2019?**; **quatraro2010?**). The coherence metric also operationalizes the “knowledge coherence” and “cognitive proximity” concepts from innovation literature (Neffke, Henning, and Boschma 2011; Boschma 2015) using our directional network structure. It enables testing whether regional technological potential (predicted feasibility) depends on the alignment in knowledge infrastructure between regional portfolios and technology network positions.

For each region r , technology i , and IPC category c , we construct two embeddedness vectors capturing i 's directional network position and compare them to r 's average embeddedness for technologies in category c .

Technology embeddedness (individual technology i):

- Incoming: $\text{embc_to}_{i,c} = \frac{\sum_{t \in c} W_{t \rightarrow i}}{|\{t \in c: W_{t \rightarrow i} > 0\}|}$ (average FITS weight from category c to technology i)
- Outgoing: $\text{embc_from}_{i,c} = \frac{\sum_{t' \in c} W_{i \rightarrow t'}}{|\{t' \in c: W_{i \rightarrow t'} > 0\}|}$ (average FITS weight from technology i to category c)

Regional average embeddedness (region r , category c):

- Incoming: $\overline{\text{embcat_to}}_{r,c} = \frac{1}{|S_{r,c}|} \sum_{t \in S_{r,c}} \text{embcat_to}_{t,c}$ where $S_{r,c} = \{t \in c : \text{RTA}_{r,t,y} \geq 1\}$
- Outgoing: $\overline{\text{embcat_from}}_{r,c} = \frac{1}{|S_{r,c}|} \sum_{t \in S_{r,c}} \text{embcat_from}_{t,c}$

Coherence is the cosine similarity between technology i 's directional embeddedness and region r 's average directional embeddedness for all categories.

$$\text{Coherence}_{r,i} = \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \cdot \|\mathbf{v}_2\|}$$

where:

- $\mathbf{v}_1 = [\text{embcat_to}_{i,c}, \overline{\text{embcat_to}}_{r,c}]$
- $\mathbf{v}_2 = [\text{embcat_from}_{i,c}, \overline{\text{embcat_from}}_{r,c}]$

Conceptually, this is close to how coherent diversification measures compute “how surrounded a technology is” by related portfolio elements, but with two key differences. First, we ground proximity in predictive, directed dependencies (FITS) rather than symmetric co-occurrence similarity. Second, we compute coherence at an internal domain scale (IPC section) to keep the measure meaningful for regional analysis under known scale heterogeneity.

In here, our coherence measure ranges from 0 to 1. In our context, high coherence (≈ 1) means that for technology i 's embeddedness signal in different categories/industries closely matches the average industrial embeddedness of region r 's portfolio of technologies. The

region’s existing capabilities align with the structural prerequisites and consequences of technology i . Alternatively, a neutral coherence (≈ 0) reflects a misalignment in industrial embeddedness between technology i ’s and the region.

Coherence serves two roles in this work. First, as dyadic fit measure: it is a direct indicator of portfolio–technology alignment in a given domain or industry, capturing whether predicted feasibility is supported by an internally coherent knowledge infrastructure. Second, as moderation channel: it allows us to test whether the effect of technology network position (e.g., bridging technologies, enabling technologies) depends on whether regions have the industry-level knowledge infrastructure to exploit those positions. This is an explicitly contextual mechanism that simple diversification counts cannot reveal (Quattraro 2010; Boschma 2015) and aligns with core idea of this work.

3.7 Summary

Our approach replaces traditional relatedness constructs with machine learning-derived measures that enable contextualized diversification analysis:

- **Technological Potential** ($p_{r,t,y}$): Region-specific, non-linear, time-varying probabilities replace linear relatedness density
- **FITS Network** ($W_{t \rightarrow t'}$): Asymmetric, predictive dependencies replace symmetric co-occurrence-based relatedness

- **Coherence** ($\text{Coherence}_{r,t,c,y}$): Structural alignment between regional portfolios and technology network positions captures knowledge infrastructure quality

Together, these measures allow testing how diversification is contingent on regional knowledge infrastructure (RQ2), national ecosystem characteristics (RQ3), and spatial factors (RQ4) in ways that traditional relatedness frameworks cannot—addressing the core problem of contextualizing diversification strategies beyond path dependency identification.

4 Empirical Design

The preceding chapters established that regional technological diversification requires contextual understanding that mainstream relatedness measures cannot provide. Our methodological contribution: Regional Technological Potential, Feature Importance Technology Space, and the Coherence measure, captures endogenous context: the complex interactions between regional technology portfolios and the technology characteristics. We also highlighted why these measures are useful conceptually, and argued how traditional linear density measures obscure such complex interactions. However, endogenous context alone is insufficient. Regional economies are part of national institutional environments as well as spatial networks that could shape which opportunities translate into diversification possibilities.

This chapter extends our framework to incorporate exogenous context. We examine how national entrepreneurial ecosystems contribute to regional potential, and investigate whether regional potential exhibits spatial dependence—do neighbouring regions’ outcomes and characteristics influence focal region possibilities?

Our empirical strategy addresses four interconnected elements:

1. **Technology-region linkages:** How network position in FITS affects potential
2. **Knowledge infrastructure alignment:** How coherence moderates these effects
3. **National context:** How entrepreneurial ecosystem characteristics shape regional possibilities
4. **Spatial dynamics:** How neighbouring regions influence focal region outcomes

These elements correspond directly to our research questions and guide the two complementary analyses that follow. These analyses are designed to explain variation in regional potential (predicted feasibility) rather than realized entry outcomes. However, we also compare how potential and relatedness density contribute to the entry outcome as a secondary benchmark. Throughout, our focus is on the capability-based opportunity structure implied by regional portfolios and technological requirements, not on observed diversification decisions. Albeit this focus might limit the interpretability of our results and we discuss this in the subsequent chapters.

4.1 Conceptual Framework

4.1.1 The Structure of Context

The relationship between regions and technologies operates through distinct yet interconnected characteristics spanning multiple analytical levels.

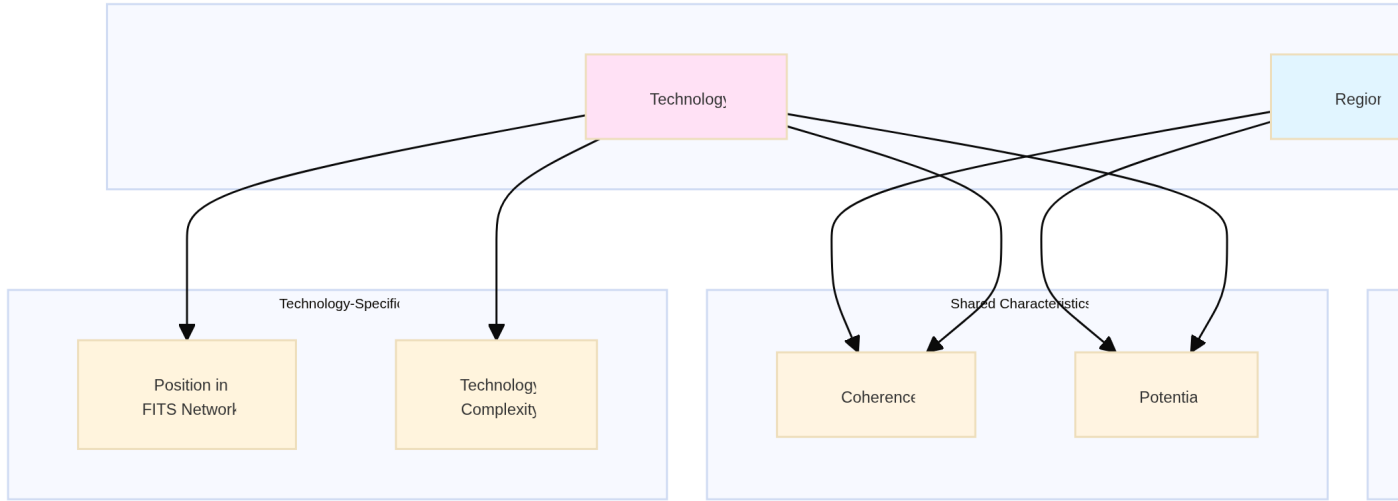
At the **technology level**, each technology occupies a position within the FITS network. Some bridge otherwise disconnected domains, functioning as general purpose technologies

(high betweenness). Others receive predictive importance from many technologies, representing key enabling technologies that multiple domains depend upon (high authority). Still others predict many downstream technologies, serving as foundational capabilities that enable subsequent development (high hub). These structural positions represent different types of opportunities with different capability requirements.

At the **region level**, each region possesses a technological portfolio characterised by its breadth, complexity, and network positioning. Regions also exist within national institutional environments and spatial neighborhoods that provide the broader context for capability development.

The critical intersection occurs at the **region-technology dyad**. Here, two properties emerge from the interaction between regional capabilities and technological requirements:

- **Coherence:** The alignment between a technology’s FITS embeddedness and the region’s portfolio embeddedness, i.e: whether the region possesses complementary knowledge infrastructure for that specific technology.
- **Potential:** The estimated probability of future specialisation ($RTA - 1$) that a region will have in a given technology. The potential a region has in a technology can be considered as metric of feasibility.



This dyadic relationship is nested within broader hierarchies. Spatially, neighbouring regions’ characteristics create spillover effects through knowledge diffusion and demonstration. Nationally, the entrepreneurial ecosystem provides institutional conditions that affect all regions within a country.

Crucially, coherence serves dual roles: as a direct predictor of potential and as a moderator of network position effects. Technologies occupying central network positions yield differential regional impacts depending on industrial alignment—high coherence regions can leverage well-connected technologies, while low coherence regions may struggle despite favourable network positions. This moderation captures the contextual contingency at the heart of our argument.

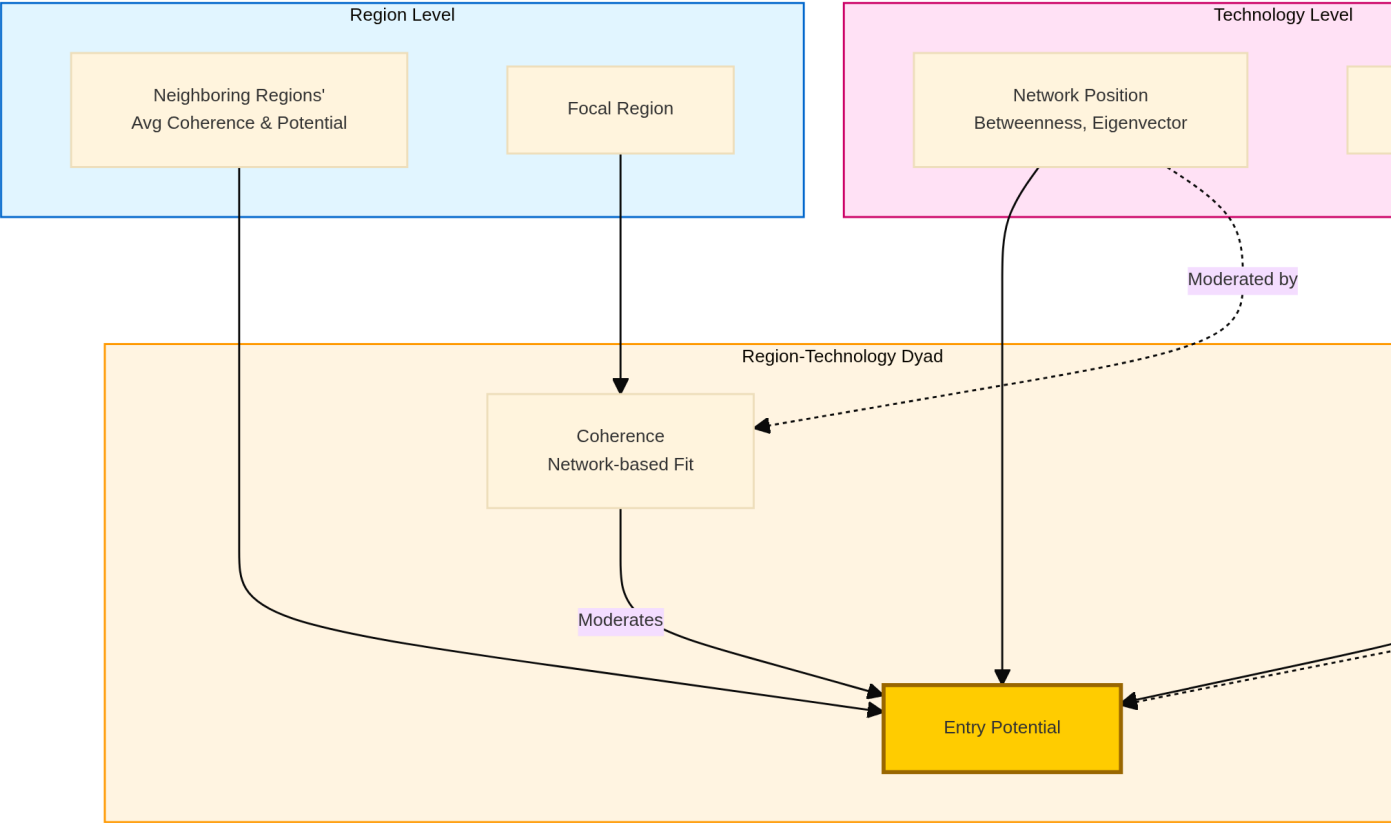
4.1.2 From Concepts to Analysis

This conceptual structure implies two complementary analyses operating at different aggregation levels:

Analysis 1 examines technology-region-year observations to test how network position, coherence, and national context shape potential. This addresses H2 (coherence moderates network effects) and H3 (national ecosystems affect potential).

Analysis 2 examines region-year panels to test whether regional potential exhibits spatial dependence beyond what regional characteristics explain. This addresses H4 (spatial spillovers of outcomes and characteristics).

Both analyses take potential as outcome but address distinct mechanisms. Analysis 1 isolates technological and institutional determinants at the dyadic level. Analysis 2 captures geographic spillovers that the technology-level analysis—which treats regions as independent—cannot identify.



Throughout, we explain variation in predicted feasibility rather than observed entry. Potential captures what regional portfolios enable given technological requirements; actual entry conflates capability with strategic choice, market conditions, and stochastic factors. By focusing on regional potential, we isolate the capability-based opportunity structure that precedes and constrains observed diversification decisions which policy can address.

4.2 Data and Variables

Our dataset comprises European patent applications from the EPO/PATSTAT database, covering 641 technologies (4-digit IPC classes) across 345 NUTS2 regions in 30 European countries over 11 years (2008–2018). After filtering for data availability, Analysis 1 includes approximately 1.25 million technology-region-year observations. Analysis 2 aggregates to approximately 3,800 region-year observations.

Our main outcome is the Regional Technological Potential $p_{r,t,y}$, which is the Random Forest-predicted probability that region r develops specialisation in technology t by year y , given its portfolio four years prior. As developed in the methodology chapter, this measure captures non-linear, region-specific diversification feasibility. Values are bounded between 0 and 1, with higher values indicating greater predicted likelihood of future specialisation.

For Analysis 2, we aggregate our outcome to regional means:

$$P_{r,y} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} p_{r,t,y}$$

We organise variables by analytical level:

Technology-region level (dyadic):

Variable	Description
Patents	patent count in technology t for region r
Coherence	Cosine similarity between technology's FITS embeddedness and region's portfolio embeddedness; interpreted as alignment between regional knowledge infrastructure and technology requirements
Distance	Shortest network path from any technology in which the region specialises to the focal technology; captures region-specific accessibility

Technology level:

Variable	Description	Interpretation
Betweenness	Betweenness centrality in FITS	General purpose technologies bridging domains
Authority	Authority score (HITS algorithm)	Key enabling technologies predicted by many others
Hub	Hub score (HITS algorithm)	Foundational technologies that predict many others

Variable	Description	Interpretation
Category	IPC section (A–H)	Technology domain

These three centrality measures exhaust the main structural roles in directed networks: bridging (betweenness), receiving importance (authority), and conferring importance (hub).

Country level:

Variable	GEI Component	Description
x8	Human capital	Education and training for entrepreneurship
x9	Competition	Market dynamics and competitive intensity
x10	Product innovation	New product development capacity
x11	Process innovation	Technology absorption and adaptation
x12	High growth	Scaling and growth orientation
x13	Internationalisation	Export orientation and global integration

Variable	GEI Component	Description
x14	Risk capital	Venture financing availability

We incorporate Global Entrepreneurship Index components capturing institutional and resource dimensions of national ecosystems. Data availability constrains our selection to components x8–x14, which span human capital, competitive intensity, innovation orientation, and financing conditions—factors most directly relevant to regional capability development and technology adoption.

Region level (for Analysis 2):

Variable	Definition
Average coherence	Average coherence across technologies
Average betweenness	Average betweenness of the region’s portfolio
Average authority	Average authority score of the region’s portfolio
Average hub	Average hub centrality of the region’s portfolio

4.3 Technology-Region Determinants of Potential

We examine what explains variation in Technological Potential across region-technology-year observations. Specifically: Do FITS network characteristics predict potential? Does

Table 4.5

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
year	11	0	2013.00	3.16	2008.00	2013.00	2018.00
pred1	1619198	0	0.18	0.21	0.00	0.09	1.00
patent_n	593	0	1.20	12.59	0.00	0.00	2377.00
coherence	1985135	0	0.55	0.22	0.00	0.57	1.00
betweenness	3156	0	0.00	0.00	0.00	0.00	0.05
authority	7039	0	0.34	0.19	0.00	0.29	1.00
hub	5529	0	0.06	0.12	0.00	0.01	1.00
degree_in	7046	0	0.65	0.16	0.00	0.64	1.00
degree_out	5534	0	0.11	0.18	0.00	0.02	1.00
degree_total	7047	0	0.20	0.15	0.00	0.13	1.00
eigenvector	7051	0	0.10	0.16	0.00	0.04	1.00
rawdist	5637	4	132.61	242.27	26.46	90.94	2883.96
relatedness_density	5537	4	9.42	8.04	0.00	7.81	100.00
reg_comp	3087	4	-0.00	1.00	-12.98	0.08	5.44
tech_comp	6833	4	-0.00	1.00	-23.15	0.01	4.63
entry	3	18	0.06	0.23	0.00	0.00	1.00
x8_human_capital	200	21	0.51	0.19	0.14	0.48	1.00
x9_competition	192	21	0.66	0.25	0.17	0.76	1.00
x10_product_innovation	196	21	0.58	0.21	0.06	0.60	1.00
x11_process_innovation	199	21	0.67	0.18	0.22	0.67	1.00
x12_high_growth	205	21	0.51	0.18	0.12	0.53	0.95
x13_internationalization	186	21	0.67	0.19	0.23	0.70	1.00
x14_risk_capital	193	21	0.66	0.16	0.19	0.67	1.00
CC.EST	330	0	1.17	0.77	-0.38	1.46	2.44
GE.EST	330	0	1.20	0.57	-0.36	1.42	2.24
PV.EST	330	0	0.62	0.40	-0.47	0.60	1.51
RQ.EST	330	0	1.27	0.45	0.14	1.27	2.04
RL.EST	330	0	1.24	0.59	-0.16	1.44	2.12
VA.EST	330	0	1.18	0.31	0.31	1.25	1.74

Table 4.6

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
year	11	0	2013.00	3.16	2008.00	2013.00	2018.00
reg_potential	1565	0	0.22	0.23	0.00	0.12	0.76
patent_count	1249	0	776.19	1985.54	0.00	186.00	27222.00
n_techs	368	0	90.03	92.69	0.00	59.00	496.00
n_specialized	210	0	59.81	49.01	0.00	50.00	241.00
diversity0	367	0	3.70	1.58	0.00	4.08	6.21
diversity	2851	0	3.21	1.33	0.00	3.58	5.28
avg_coherence	3097	0	0.55	0.04	0.11	0.56	0.64
avg_tech_comp	11	0	-0.00	0.00	-0.00	0.00	0.00
entry_rate	1946	0	0.06	0.05	0.00	0.05	0.26
avg_rd	3083	0	9.42	7.74	0.00	7.97	38.71
avg_dist	11	0	132.61	13.50	107.08	137.50	151.91
avg_betweenness	3080	0	0.00	0.00	0.00	0.00	0.04
avg_eigenvector	3087	0	0.07	0.04	0.00	0.06	0.52
med_betweenness_spec	2106	0	0.00	0.00	0.00	0.00	0.04
med_eigenvector_spec	2554	0	0.03	0.03	0.00	0.02	0.52
med_betweenness_exist	1925	0	0.00	0.00	0.00	0.00	0.04
med_eigenvector_exist	2455	0	0.03	0.03	0.00	0.02	0.52
pct_high_between_spec	124	0	0.46	0.12	0.00	0.50	0.50
pct_high_eigen_spec	109	0	0.46	0.12	0.00	0.50	0.50
pct_high_between_exist	197	0	0.46	0.12	0.00	0.50	0.50
pct_high_eigen_exist	183	0	0.46	0.12	0.00	0.50	0.50

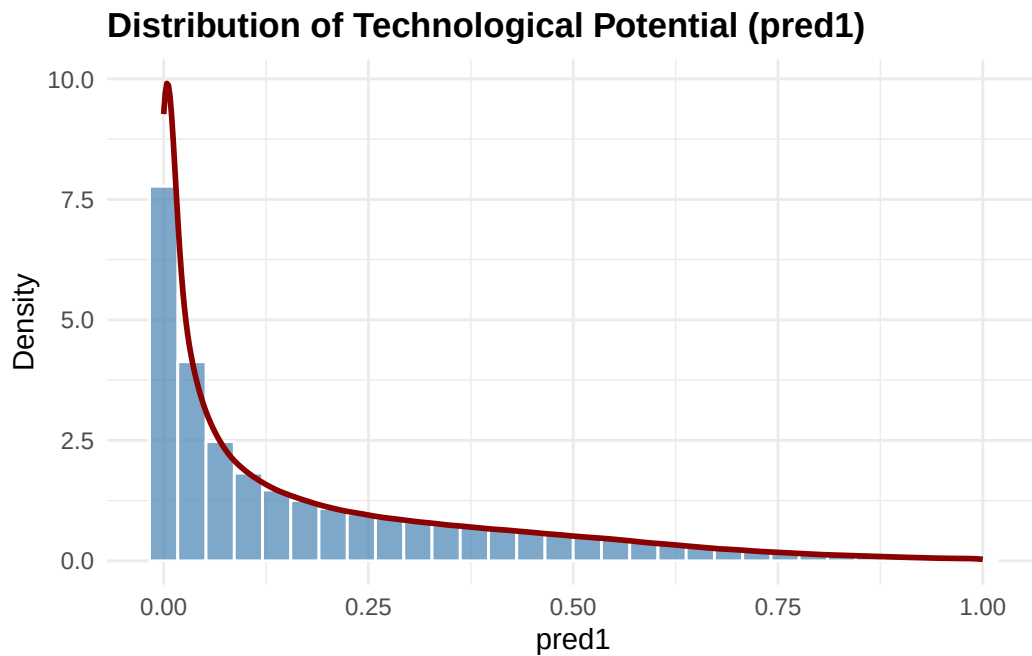


Figure 4.1

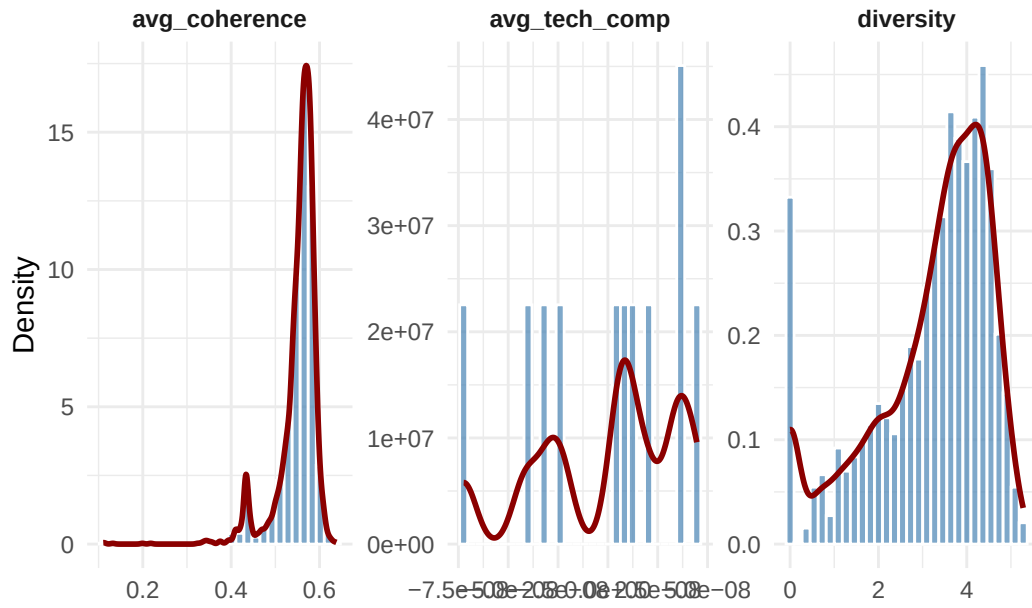


Figure 4.2

Table 4.7

	pred1	rawdist	patent_n	coherence	betweenness	eigenvector	authority
pred1	1	.	.	.	.	.	.
rawdist	-.07	1	.	.	.	.	.
patent_n	.24	-.02	1	.	.	.	.
coherence	.40	-.19	.11	1	.	.	.
betweenness	.23	-.10	.05	.40	1	.	.
eigenvector	-.15	.08	-.02	-.21	-.17	1	.
authority	-.13	-.03	-.06	-.18	-.25	-.06	1
hub	.20	-.06	.12	.33	.15	-.06	.10
relatedness_density	.69	-.00	.22	.12	.02	-.02	-.01
reg_comp	.34	.00	.09	.03	-.01	.01	.01
tech_comp	-.20	.05	-.11	-.28	-.24	-.01	-.02
entry	.29	-.02	.26	.13	.07	-.05	-.05

Table 4.8

	reg_potential	patent_count	n_techs	diversity	avg_coherence	avg_betweenness
reg_potential	1	.	.	.	.	.
patent_count	.51	1	.	.	.	.
n_techs	.91	.75	1	.	.	.
diversity	.74	.37	.74	1	.	.
avg_coherence	.52	.30	.54	.77	1	.
avg_betweenness	-.05	-.09	-.06	.13	.20	1
avg_eigenvector	.20	.17	.23	.27	.28	.
entry_rate	.76	.39	.78	.78	.52	.
avg_rd	.91	.58	.95	.85	.59	.

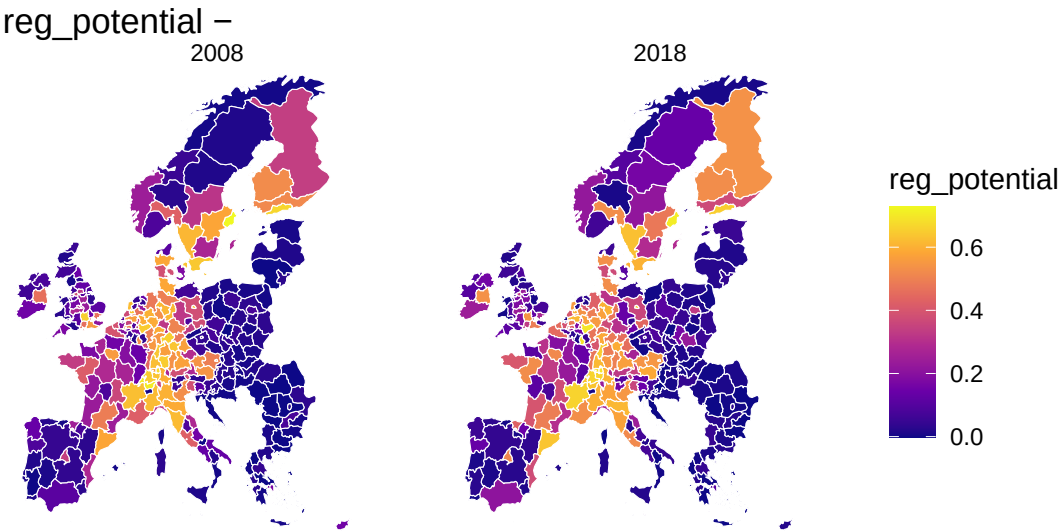


Figure 4.3

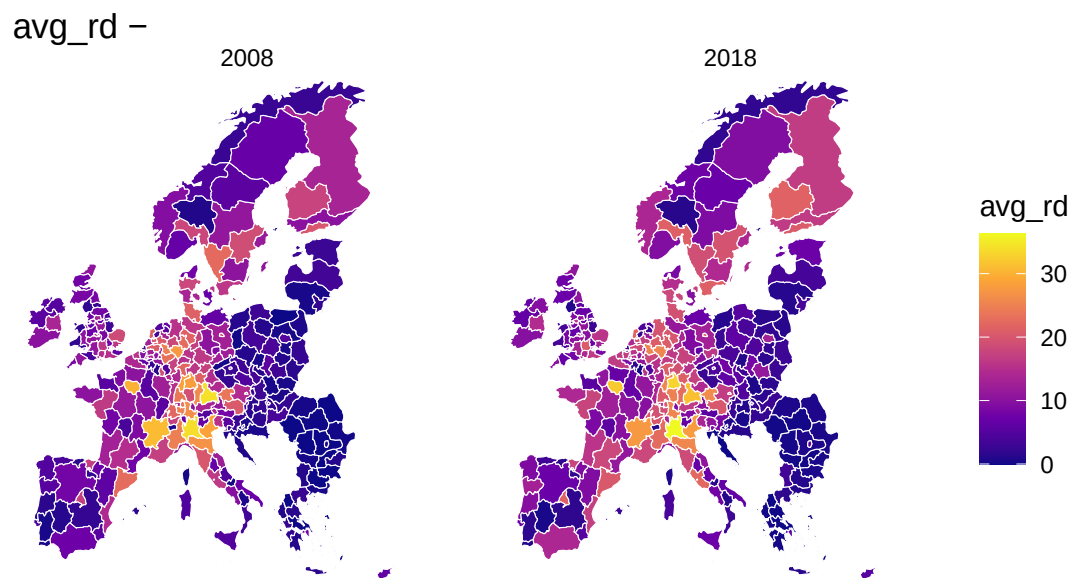


Figure 4.4

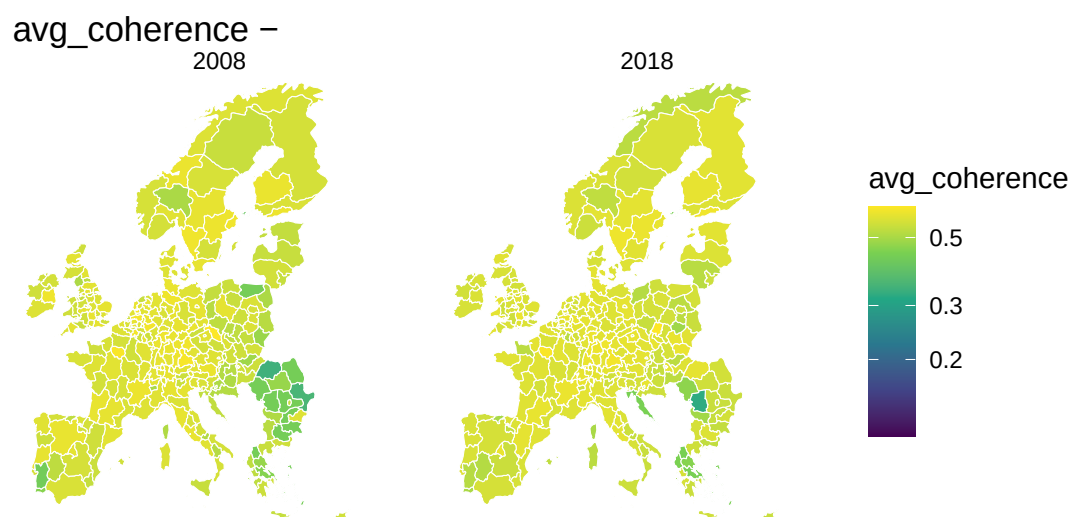


Figure 4.5

coherence moderate these relationships? Do national ecosystem factors contribute beyond regional and technological characteristics?

We estimate hierarchical models that account for the nested structure of observations. Technologies differ in inherent characteristics. Regions differ in portfolio composition. Countries differ in institutional environments. Years capture common temporal dynamics. Failing to account for this structure would conflate within-group and between-group variation.

We adopt Generalized Additive Models (GAM) with beta-distributed errors. Beta regression is appropriate because potential is bounded between 0 and 1 with continuous density—standard linear regression would produce predictions outside this range and mischaracterise the error distribution.

Random effects at each level absorb unobserved heterogeneity while allowing estimation of level-specific predictors. We specify random effects for year, technology, and region. A fully nested specification (region within country) would better isolate national-level GEI effects but proved computationally infeasible given our sample size of 1.25 million observations. The regional random effect therefore captures combined geographic variation, and GEI coefficients should be interpreted as associations conditional on this heterogeneity.

We estimate two complementary specifications consistent with our pipeline: (i) fixed-effects dyad models as a robustness check (region-technology (rt) dyad and year fixed effects), and (ii) hierarchical beta-GAM models as our primary approach, estimated with random-effect smooths.

The general form is:

$$g(\mu_{r,t,y}) = \mathbf{X}_{r,t,y}\beta + f_{\text{year}} + f_{\text{region}} + f_{\text{tech}}$$

where $g(\cdot)$ is the logit link function, $\mu_{r,t,y} = E[\text{Potential}_{r,t,y}]$, and f are random effect smooths implemented as penalised ridge regression (equivalent to random intercepts in mixed models).

We estimate two complementary beta-GAM specifications:

- (A) Interaction model (tests H2): includes patents, distance, regional and technological complexity, and interaction terms between FITS network position and coherence (betweenness $\times$ coherence, authority $\times$ coherence, hub $\times$ coherence), with random effects for year, region, and technology.
- (B) Hierarchical model with national context (tests H3): adds GEI components (x8–x14) and category controls, and includes a country random effect, without interaction terms.

Main effects:

- $\beta_1 \text{patents}_{r,t,y}$: Regional knowledge stock in focal technology
- $\beta_2 \text{coherence}_{r,t,y}$: Portfolio-technology alignment
- $\beta_3 \text{distance}_{r,t,y}$: Network accessibility from regional portfolio
- $\beta_4 \text{betweenness}_t$: Bridging position (GPT)
- $\beta_5 \text{authority}_t$: Enabling technology status (KET)

- $\beta_6 \text{hub}_t$: Foundational technology status

Interactions (testing H2):

- $\beta_7(\text{coherence} \times \text{betweenness})$
- $\beta_8(\text{coherence} \times \text{authority})$
- $\beta_9(\text{coherence} \times \text{hub})$
- $\beta_{10}(\text{coherence} \times \text{distance})$

National context (testing H3):

- $\sum_{k=8}^{14} \delta_k \text{GEI}_k$

Controls:

- $\gamma_c \text{category}_t$: Technology domain fixed effects

Random effects:

- $f_{\text{year}}, f_{\text{region}}, f_{\text{tech}}$

Estimation uses `bam()` from the `mgcv` package, which implements discretised covariate methods for computational efficiency with large datasets.

The interaction terms test H2 directly. A positive coefficient on $\text{coherence} \times \text{betweenness}$ indicates that bridging technologies (GPTs) yield higher potential specifically when regional portfolios align with the bridged domains. This supports the contextual argument: structural opportunity requires complementary capabilities. Similar interpretation applies to authority and hub interactions.

The GEI coefficients test H3. Positive effects indicate that national ecosystem characteristics enable higher regional potential, conditional on technological and portfolio factors. For instance, a positive coefficient on human capital suggests that regions in countries with stronger educational infrastructure exhibit higher potential for given technology-portfolio configurations.

4.4 Spatial Dynamics of Regional Potential

We shift from technology-region observations to region-year panels to examine whether regional potential exhibits spatial structure. The technology-level analysis treats regions as independent observations. But regions exist in geographic space where knowledge diffuses, competition operates, and demonstration effects propagate. Analysis 2 tests whether potential clusters geographically and whether neighbors' outcomes and characteristics influence focal region possibilities.

Before specifying spatial models, we verify that spatial structure exists. Moran's I statistics reveal persistent positive spatial autocorrelation across all years ($I = 0.44\text{--}0.51$, $p < 0.001$). Regional potential clusters geographically—high-potential regions systematically neighbor high-potential regions.

To discriminate between competing spatial specifications, we apply Baltagi-Song-Koh LM tests designed for panel data.

The robust LM-lag test (350.4, $p < 0.001$) strongly rejected the null of no spatial lag dependence. The robust LM-error test (4.1, $p = 0.042$) was only marginally significant

statistic	p.value	parameter	method
616.271281	4.838869e-136	1	LM test for spatial lag dependence
271.667219	4.915301e-61	1	LM test for spatial error dependence
348.251774	1.018224e-77	1	Locally robust LM test for spatial lag dependence sub spatia
3.647713	5.614626e-02	1	Locally robust LM test for spatial error dependence sub spat

after controlling for spatial lag. This indicates that spatial dependence operates primarily through neighbors' outcomes rather than through spatially correlated errors—regions with high-potential neighbors tend to exhibit higher potential themselves.

Based on these diagnostics, we adopt a Spatial Autoregressive (SAR) panel model with random effects as our primary specification.

We construct spatial weights using k-nearest neighbors ($k = 5$) based on geographic centroids:

$$w_{ij} = \begin{cases} 1/k & \text{if } j \in \text{knn}_k(i) \\ 0 & \text{otherwise} \end{cases}$$

with row-standardisation such that $\sum_j w_{ij} = 1$ for each region i .

We estimate three specifications to isolate different spatial mechanisms:

SAR (testing H4a):

$$P_{r,y} = \rho \mathbf{W} P_{r,y} + \mathbf{X}_{r,y} \beta + \alpha_r + \epsilon_{r,y}$$

Tests whether neighbors' potential affects focal region (outcome spillovers). A significant

$\rho > 0$ indicates that high-potential neighbors are associated with higher focal region potential, suggesting knowledge spillovers or demonstration effects.

SEM (robustness):

$$P_{r,y} = \mathbf{X}P_{r,y}\beta + \alpha_r + uP_{r,y}, \quad u_{r,y} = \delta \mathbf{W}u_{r,y} + \epsilon_{r,y}$$

where δ captures spatial correlation in the error term (unobserved spatially correlated shocks).

The spatial models test H4 directly by examining whether regional technological potential is influenced by neighboring regions' outcomes and characteristics.

In the SAR specification, the spatial lag coefficient ρ captures outcome spillovers. A positive and significant ρ indicates that regions tend to exhibit higher potential when neighboring regions also display high potential. This is consistent with knowledge diffusion, demonstration effects, and interregional learning, implying that capability-based opportunity structures are not purely local but propagate across space (H4a).

Regional controls (technological complexity and patenting intensity) ensure that spatial dependence is not driven by simple similarities in regional size or diversification. When ρ remains significant conditional on these controls, spatial interdependence reflects genuine spillovers rather than compositional effects.

The SEM specification tests an alternative mechanism. A significant spatial error parameter δ (sometimes also referred to as λ) would indicate that similarities in regional potential arise from unobserved spatially correlated factors, such as shared institutional

environments or common shocks, rather than direct interaction between neighbouring regions.

Comparing SAR and SEM results allows us to distinguish between interactive spillovers (neighbours' outcomes matter) and correlated environments (neighbors share unobserved conditions).

5 Results

Our empirical analysis proceeds in five stages. First, we benchmark how much standard relatedness density and our RF-based Potential explain next-period entry under progressively stricter comparisons (Table Table 5.1). Second, we examine how technology network positions, regional knowledge infrastructure, and accessibility shape diversification potential using region and year fixed effects (Table Table 5.2). Third, we incorporate national entrepreneurial ecosystem characteristics to test whether these dyadic mechanisms persist once national institutions are accounted for (Table Table 5.3). Fourth, we estimate hierarchical random effects models to reflect the nesting of regions within countries and to assess how much variation lies between versus within regional units (Table Table 5.4). Fifth, we aggregate potential to the regional level and estimate spatial panel models to test whether regional potential exhibits spatial dependence consistent with spillovers or correlated unobservables (Tables Table 5.5 and Table 5.6). Together, these analyses test our hypotheses while checking whether conclusions depend on model structure or level of aggregation.

5.1 Entry

In Table 5.1 we benchmark how much standard relatedness density and our RF-based Potential (pred11) explain next-period entry when we progressively tighten comparisons. In the pooled specifications, both measures strongly predict entry: relatedness density is positive and precisely estimated (0.055), while Potential is larger (0.081). This is consistent with the interpretation that both capture opportunity, while Potential is closer to feasibility as learned from portfolio configurations. Once we move to the fixed-effects designs and absorb region and year fixed effects, interpretation shifts to within-region changes over time, net of common year shocks. In this setting, relatedness density remains strongly positive (e.g., 0.089 in the density-only fixed effects model), and Potential also remains strongly positive (0.089 in the Potential-only fixed effects model), indicating that both track time-varying entry propensity beyond stable regional differences.

The most informative column is the joint fixed effects model, where both predictors enter together. Here, relatedness density drops from 0.089 to 0.028, while Potential remains large at 0.086. This pattern implies that Potential subsumes a substantial share of what density captures, but not all of it: density retains an independent association with entry even after controlling for Potential. In other words, Potential appears to be the dominant predictor in this horse race, while density retains residual explanatory content that may reflect a simpler proximity component that survives beyond the non-linear feasibility signal embedded in Potential. This dominance is also reflected in fit: within- R^2 rises from 0.014 in the density-only fixed effects model to 0.062 in the Potential-only fixed effects model and remains 0.062 in the joint model. This benchmark motivates the subsequent mechanism

analysis: rather than treating relatedness as a sufficient summary statistic, we examine how portfolio alignment, network position, and accessibility structure feasibility. This move is consistent with the broader methodological critique that pairwise relatedness can miss higher-order portfolio information; for example, (**RelatednessMachineLearning?**) argues that shifting from “two-product correlations” to information that captures richer configurations can improve predictive performance. This benchmark motivates our subsequent mechanism analysis: having established that Potential dominates relatedness density in predicting entry, we now decompose Potential itself to examine how portfolio alignment (coherence), network position (betweenness, eigenvector), and accessibility structure feasibility at the dyadic technology-region level

5.2 Network Effects and Coherence Moderation

Table 5.2 presents our core test of how technology network positions and regional knowledge infrastructure shape diversification potential. Column 4, our preferred specification with region and year fixed effects, shows that coherence—the alignment between a technology’s FITS embeddedness and the region’s portfolio—strongly predicts potential (0.706). This aligns with the capabilities-based interpretation of relatedness in evolutionary economic geography. (**TechNetworkResilience?**) states that “economic activities are related through sharing various capabilities,” and it links diversification to learning enabled by “cognitive proximity and interactive learning” following Boschma’s proximity framework. In our setting, coherence operationalizes this shared-capability, portfolio-fit mechanism and remains stable across specifications, consistent with its interpretation as

Table 5.1

	baseline1	baseline2	full1	full2
(Intercept)	0.069*** (0.002)	0.070*** (0.001)		
patent_n	0.008 (0.007)	-0.007** (0.002)	0.017* (0.008)	0.004 (0.003)
relatedness_density	0.055*** (0.002)		0.089*** (0.006)	
reg_comp	-0.003*** (0.001)	-0.003*** (0.001)	-0.001*** (0.000)	-0.000 (0.000)
tech_comp	-0.015*** (0.002)	-0.005*** (0.001)	-0.013*** (0.002)	-0.004*** (0.001)
pred11		0.081*** (0.002)		0.089*** (0.002)
Num.Obs.	1571364	1571364	1571364	1571364
R2	0.043	0.086	0.046	0.092
R2 Adj.	0.043	0.086	0.046	0.092
R2 Within			0.014	0.062
R2 Within Adj.			0.014	0.062
AIC	-36976.1	-109812.6	-41129.6	-119089.6
BIC	-36914.8	-109751.3	-37375.8	-115335.8
RMSE	0.24	0.23	0.24	0.23
Std.Errors	by: region & tech	by: region & tech	by: region & tech	by: region & tech
FE: region			X	X
FE: year			X	X
FE: tech				

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

a structural feasibility correlate rather than a modeling artifact.

Betweenness centrality, capturing bridge positions that connect domains, exhibits substantial positive effects (0.388). However, the coherence $\times$ betweenness interaction is significantly negative (-0.284), revealing contextual contingency: bridging positions disproportionately benefit regions with lower coherence, offering catch-up opportunities when portfolios are misaligned with these bridge technologies. Conversely, high-coherence regions gain less from bridging positions, suggesting that their feasible opportunities are less dependent on bridge roles and more concentrated in already-aligned portions of the technology space. This contingency is conceptually consistent with a tension highlighted in (**TechNetworkResilience?**). After reviewing the standard logic that related variety supports diversification, it notes that interdependencies can also transmit shocks and that evidence on how internal relatedness structures shape outcomes is mixed; it concludes that it is “still unclear” how relatedness shapes resilience and that “networks are of great assistance” because economies are “webs” of interdependent knowledge and activities. Our interaction result is consistent with that network-conditional view: the value of a network position depends on portfolio context rather than being uniform.

The eigenvector centrality pattern presents a contrasting story. Technologies occupying authority positions consistently inhibit entry (e.g., -0.221 in Column 4), yet the coherence $\times$ eigenvector interaction remains non-significant. A conservative interpretation is that authority-like targets are systematically difficult to enter regardless of alignment as captured by coherence. This pattern is compatible with arguments that diversification requires more than proximity. (**GoingBeyondRelatednessKETs?**) argues that diver-

Table 5.2: logit(pred1) - ols

	baseline	network	interaction_a	interaction_a1
(Intercept)	-2.528*** (0.070)		-2.462*** (0.074)	-2.476*** (0.074)
patent_n	0.211*** (0.054)	0.088*** (0.023)	0.213*** (0.053)	0.215*** (0.054)
coherence	1.026*** (0.038)	0.779*** (0.039)	0.898*** (0.039)	0.909*** (0.040)
rawdlist	-0.028 (0.028)	-0.039 (0.028)	-0.011 (0.027)	
reg_comp	0.741*** (0.065)	0.013 (0.018)	0.746*** (0.066)	0.741*** (0.065)
tech_comp	-0.213*** (0.035)	-0.256*** (0.040)	-0.212*** (0.035)	-0.198*** (0.034)
betweenness		0.131*** (0.023)	0.249*** (0.039)	0.266*** (0.040)
eigenvector		-0.237*** (0.030)	-0.206*** (0.028)	-0.201*** (0.029)
betweenness × coherence			-0.184*** (0.032)	-0.191*** (0.033)
coherence × eigenvector			-0.015 (0.030)	-0.001 (0.031)
betweenness × rawdlist				
rawdlist × eigenvector				
Num.Obs.	1906137	1906137	1906137	1922371
R2	0.354	0.659	0.366	0.367
R2 Adj.	0.354	0.659	0.366	0.367
R2 Within		0.356		
R2 Within Adj.		0.356		
AIC	7682716.6	6466221.8	7645396.3	7738488.6
BIC	7682791.4	6470084.6	7645520.9	7738600.8
RMSE	1.82	1.32	1.80	1.81
Std.Errors	by: region & tech	by: region & tech	by: region & tech	by: region & tech
FE: year		X		
FE: region		X		

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

sification is “going beyond relatedness” and reports that enabling technologies support more exploratory trajectories “only when” they are used by other local technologies, implying a complementarity and integration condition beyond portfolio fit alone. Our null moderation for eigenvector is consistent with the idea that barriers for authority targets reflect complements or coordination not captured by alignment measures.

Distance becomes most informative once interactions are introduced. In Table Table 5.2, the main effect of rawdist is not stable in the fixed effects setting, but the interaction between rawdist and eigenvector is negative and robust (around -0.080 in the distance-interaction specifications). This indicates that authority technologies become particularly difficult when they are also topologically remote from any regional foothold. This mechanism is close to the account in (**StrongLinksWeakLinksUnrelatedSurvive?**), which describes network links as “pipelines” and states that “the strength of links reflects technological distance,” with the implication that predictable transfer is more feasible through stronger, closer links. In contrast, the betweenness $\times$ rawdist interaction is small and not robust in the fixed effects specifications (e.g., 0.067 and 0.057 with large standard errors), suggesting limited evidence in within-region comparisons that bridge positions systematically enable longer jumps purely through remoteness.

5.3 National Context and Robustness

Table 5.3 extends our framework by incorporating national entrepreneurial ecosystem characteristics from the Global Entrepreneurship Index. Column 4 shows that the core

dyadic mechanisms persist: coherence remains positive (0.709), betweenness remains positive (0.398), and the coherence $\times$ betweenness interaction remains negative (-0.292). The eigenvector penalty likewise persists (around -0.222) without coherence moderation, and the distance–authority interaction remains negative (rawdist $\times$ eigenvector around -0.082). This stability supports a cautious conclusion that the coherence and network-position mechanisms are not simply proxies for national institutional variation. At the same time, national ecosystem variables contribute additional explanatory structure. Competition is consistently positive (about 0.088 in Column 4), while internationalization is consistently negative (about -0.082). Other GEI components show heterogeneous patterns across specifications, suggesting they may operate differently in cross-sectional comparisons than in within-country temporal change. This combination of stable dyadic mechanisms and non-negligible national effects is consistent with **(RelatednessAndEconComplexityPolicy?)**, which cautions against treating relatedness metrics as sufficient guides to diversification because institutional and policy context can shape which opportunities are realized. This heterogeneity notwithstanding, the consistency of competition and internationalization effects, alongside the stable core mechanisms, provides sufficient evidence for H3 that national entrepreneurial ecosystems shape regional potential beyond dyadic technology-region characteristics.

5.4 Hierarchical Structure and Spatial Implications

Table 5.4 Table 5.4 employs random effects models that account for regions nested within countries, a specification that matches the hierarchical structure of the data and is appro-

Table 5.3: logit(pred1) - ols

	h0	h01	h1	h2
(Intercept)	-2.493*** (0.060)		-2.406*** (0.063)	
patent_n	0.181*** (0.044)	0.090*** (0.023)	0.181*** (0.045)	0.090*** (0.024)
coherence	0.884*** (0.039)	0.784*** (0.040)	0.826*** (0.038)	0.709*** (0.038)
rawdist	-0.024 (0.030)	-0.037 (0.030)	-0.015 (0.029)	-0.027 (0.029)
reg_comp	0.519*** (0.063)	0.015 (0.021)	0.518*** (0.063)	0.011 (0.020)
tech_comp	-0.215*** (0.039)	-0.243*** (0.042)	-0.214*** (0.039)	-0.241*** (0.041)
betweenness	0.104*** (0.023)	0.133*** (0.024)	0.316*** (0.038)	0.398*** (0.038)
eigenvector	-0.224*** (0.031)	-0.240*** (0.031)	-0.211*** (0.028)	-0.222*** (0.028)
x8_human_capital	-0.132* (0.054)	0.044* (0.022)	-0.134* (0.054)	0.044* (0.022)
x9_competition	0.568*** (0.082)	0.087* (0.034)	0.573*** (0.082)	0.088** (0.034)
x10_product_innovation	0.198*** (0.056)	0.016 (0.018)	0.200*** (0.056)	0.017 (0.018)
x11_process_innovation	0.235*** (0.063)	-0.020 (0.028)	0.236*** (0.063)	-0.022 (0.028)
x12_high_growth	-0.110+ (0.056)	0.008 (0.020)	-0.109+ (0.056)	0.009 (0.020)
x13_internationalization	-0.088 (0.054)	-0.080** (0.029)	-0.090 (0.054)	-0.082** (0.029)
x14_risk_capital	0.109* (0.045)	-0.035 (0.023)	0.109* (0.046)	-0.035 (0.023)
coherence × betweenness			-0.234*** (0.031)	-0.292*** (0.031)
coherence × eigenvector			-0.006 (0.030)	0.002 (0.030)
betweenness × rawdist				
rawdist × eigenvector				
Num.Obs.	1506300	1506300	1506300	1506300
R2	0.474	0.661	0.478	0.668
R2 Adj.	0.474	0.661	0.478	0.668
R2 Within		0.358		0.371
R2 Within Adj.		0.358		0.371
AIC	5748605.3	5088428.5	5735734.5	5057169.8
BIC	5748788.7	5092279.4	5735942.3	5061045.2
RMSE	1.63	1.31	1.62	1.30
Std.Errors	by: region & tech			
FE: region		X		X
FE: tech		Y		Y

priate given that the GEI variables are measured at the national level. The high intra-class correlation, $ICC = 0.60$, implies that 60 of variance in potential is between units rather than within-region over time. The variance components show that country-level heterogeneity (standard deviation approximately 1.12–1.17) exceeds region-within-country heterogeneity (approximately 1.02–1.04). These magnitudes do not contradict the fixed effects results; rather, they clarify that while the fixed effects estimates document robust within-region mechanisms, a substantial portion of the feasibility landscape is structured by relatively persistent differences across countries and regions.

Crucially, the core coefficients remain stable in the hierarchical models. In the random effects specification comparable to the fixed effects benchmark, coherence remains positive (around 0.709), betweenness remains positive (around 0.398), and the coherence $\times$ betweenness interaction remains negative (around -0.292). Eigenvector centrality remains negative (around -0.222) without coherence moderation. Distance is more salient in this setting: rawdist becomes significant and negative as a main effect (about -0.027), consistent with the idea that topological accessibility matters for cross-regional comparisons even if it is less informative for within-region shifts once fixed effects absorb time-invariant accessibility differences. At the same time, the interaction results reinforce the earlier asymmetry. The rawdist $\times$ eigenvector term remains strongly negative (around -0.082), indicating that remoteness compounds difficulty specifically for authority targets. The betweenness $\times$ rawdist interaction becomes positive and significant in the random effects model (e.g., 0.050–0.061), suggesting that bridging positions may be more relevant for overcoming remoteness in between-place comparisons than in within-region variation. Because

these patterns appear most clearly in the cross-sectional component captured by random effects, they should be interpreted as differences in accessibility across places rather than as short-run within-place responses.

These hierarchical results also help justify the spatial models that follow. First, the large *ICC* indicates that potential is strongly structured across places, not only through time-varying within-region dynamics. Second, (**TechNetworkResilience?**) emphasizes that regional economies are “webs” of interdependent activities and argues that “networks are of great assistance,” a framing that makes it plausible that feasibility is not independent across space but may be shaped by neighboring structures. Third, the distinction between within-region fixed effects patterns and between-place random effects patterns suggests that part of the remaining heterogeneity may be geographically patterned, motivating an explicit test for spatial dependence.

5.5 Role of space

Having established stable dyadic mechanisms at the technology–region level (Tables Table 5.2 and Table 5.3) and substantial between-place heterogeneity in the hierarchical models (Table Table 5.4), we test whether regional potential exhibits spatial dependence. We aggregate potential to region-year means and estimate spatial panel models with 5-nearest-neighbor weights matrices. Tables Table 5.5 and Table 5.6 reveal substantial spatial dependence. In the SAR model, the spatial autoregressive parameter is around 0.376 in the interaction specification, indicating that regional potential is positively associated

Table 5.4: logit(pred1) - ols

	inter_a	inter_al	inter_b	inter_b1	h_a	h_b
(Intercept)	-2.577*** (0.088)	-2.667*** (0.087)	-2.573*** (0.088)	-2.666*** (0.088)	-2.817*** (0.227)	-2.711*** (0.230)
patent_n	0.109*** (0.001)	0.108*** (0.001)	0.088*** (0.001)	0.088*** (0.001)	0.090*** (0.001)	0.090*** (0.001)
rawdist	-0.032*** (0.001)	0.008*** (0.002)	-0.029*** (0.001)	0.009*** (0.002)	-0.037*** (0.001)	-0.027*** (0.001)
betweenness	0.427*** (0.002)	0.200*** (0.001)	0.388*** (0.002)	0.147*** (0.001)	0.133*** (0.001)	0.398*** (0.002)
coherence	0.729*** (0.001)	0.802*** (0.001)	0.706*** (0.001)	0.780*** (0.001)	0.784*** (0.001)	0.709*** (0.001)
eigenvector	-0.212*** (0.001)	-0.223*** (0.001)	-0.221*** (0.001)	-0.233*** (0.001)	-0.240*** (0.001)	-0.222*** (0.001)
betweenness × coherence	-0.270*** (0.001)		-0.284*** (0.001)			
coherence × eigenvector	0.000 (0.001)		-0.001 (0.001)			0.002 (0.001)
betweenness × rawdist		0.061*** (0.004)		0.057*** (0.004)		
rawdist × eigenvector		-0.080*** (0.001)		-0.080*** (0.001)		
reg_comp			0.010*** (0.001)	0.014*** (0.001)	0.015*** (0.002)	0.011*** (0.002)
tech_comp			-0.254*** (0.001)	-0.255*** (0.001)	-0.243*** (0.001)	-0.241*** (0.001)
x8_human_capital					0.044*** (0.003)	0.044*** (0.003)
x9_competition					0.088*** (0.005)	0.088*** (0.005)
x10_product_innovation					0.016*** (0.002)	0.017*** (0.002)
x11_process_innovation					-0.020*** (0.004)	-0.022*** (0.004)
x12_high_growth					0.008** (0.003)	0.009** (0.003)
x13_internationalization					-0.080*** (0.003)	-0.082*** (0.003)
x14_risk_capital					-0.035*** (0.002)	-0.034*** (0.002)
coherence × betweenness						-0.292*** (0.002)
rawdist × betweenness						
SD (Observations)	1.346	1.356	1.307	1.318	1.310	1.296
SD (Intercept regioncountry)					1.029	1.036
SD (Intercept country)					1.124	1.135
SD (Intercept region)	1.497	1.489	1.513	1.499		
Num.Obs.	1999432	9999432	1906137	1906137	1506300	1506300
R2 Marg.	0.190	0.186	0.202	0.198	0.197	0.201
R2 Adj.	0.182	0.181	0.199	0.195	0.199	0.200

Table 5.5

	baseline	network	interaction
(Intercept)	0.116*** (0.003)	0.136*** (0.003)	0.132*** (0.003)
reg_comp	0.060*** (0.003)	0.055*** (0.003)	0.054*** (0.003)
patent_count	0.077*** (0.003)	0.062*** (0.003)	0.062*** (0.003)
rho	0.464*** (0.015)	0.375*** (0.016)	0.376*** (0.016)
avg_coherence		0.055*** (0.003)	0.065*** (0.003)
avg_betweenness		-0.009** (0.003)	0.009* (0.004)
avg_eigenvector		0.018*** (0.003)	0.022*** (0.003)
avg_coherence $\times$ avg_betweenness			0.013*** (0.002)
avg_coherence $\times$ avg_eigenvector			0.006*** (0.002)
Num.Obs.	3223	3223	3223
Log.Lik.	4350.51	4559.40	4578.55
AIC	-8693.02	-9104.81	-9139.09
BIC	-8668.71	-9062.26	-9084.39
Sigma	0.161	0.149	0.148
Res.Var	0.024	0.021	0.021
EDF Total	4	7	9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

with neighboring regions' outcomes. In the SEM model, the spatial error parameter is around 0.420 in the interaction specification, indicating spatial correlation in unobserved components. These estimates are consistent with the interpretation that spatial context matters for regional feasibility landscapes, whether through spillovers or through correlated shocks and unobservables.

The regional aggregate covariates also help interpret the relationship between dyadic feasibility and portfolio-level advantage. Average coherence is positive (about 0.065 in the

SAR interaction model and 0.073 in the SEM interaction model), consistent with the dyadic evidence that alignment is a central feasibility correlate. Average eigenvector centrality of existing portfolios is also positive (about 0.022 in SAR and 0.021 in SEM), and it interacts positively with average coherence (about 0.006 in SAR and 0.005 in SEM). This differs in sign from the dyadic eigenvector penalty for candidate technologies (around -0.22). A conservative reconciliation is that these statistics capture different objects: dyadic eigenvector describes the difficulty of entering an authority-like target technology, whereas average eigenvector of a region’s existing specializations reflects accumulated capabilities embedded in those technologies that may raise overall potential across the opportunity set. This distinction is compatible with the enabling-technology argument in (**GoingBeyondRelatednessKETs?**), which emphasizes that enabling capacities matter when they are embedded and used within the local system, rather than as isolated targets.

A similar distinction applies to betweenness. At the dyadic level, bridge positions are more valuable for low-coherence regions, reflected in the negative coherence $\times$ betweenness term (around -0.28). At the regional level, average betweenness is modest in isolation but becomes positive once interacted with average coherence (about 0.013 in SAR and 0.012 in SEM). This suggests that advantaged portfolios may combine alignment and bridging among what regions already do, while at the margin bridging targets provide relatively greater benefits when alignment is weak. Overall, the spatial results do not overturn the dyadic mechanisms; instead, they indicate that feasibility is also spatially structured, and that portfolio composition at the regional level can be associated with higher average

LISA Clusters – 2018

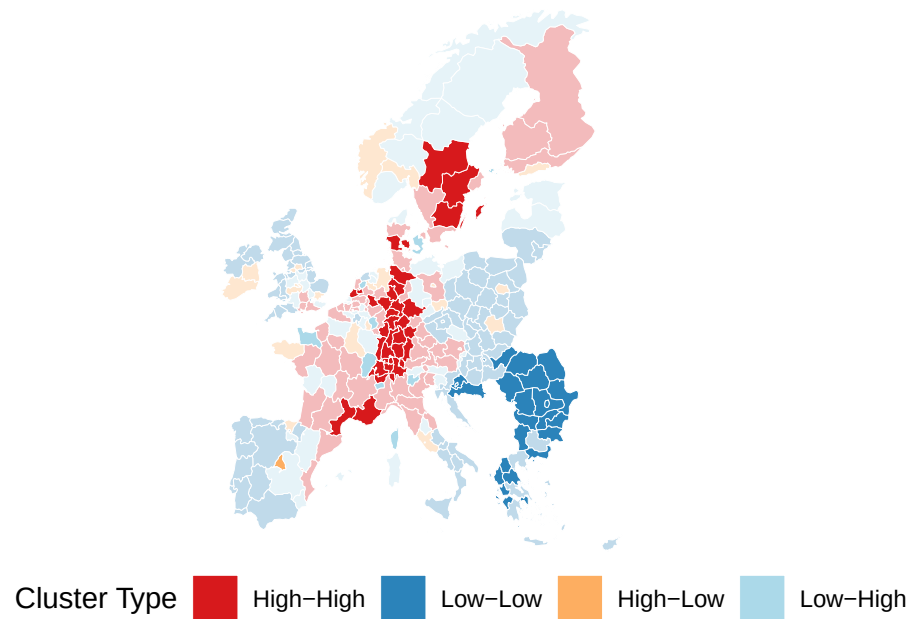


Figure 5.1

potential even when entry into specific authority targets remains difficult.

Table 5.6

	baseline	network	interaction
(Intercept)	0.217*** (0.006)	0.221*** (0.005)	0.217*** (0.005)
reg_comp	0.060*** (0.003)	0.059*** (0.003)	0.058*** (0.003)
patent_count	0.072*** (0.003)	0.059*** (0.003)	0.059*** (0.003)
delta	0.522*** (0.019)	0.421*** (0.023)	0.420*** (0.023)
avg_coherence		0.063*** (0.003)	0.073*** (0.004)
avg_betweenness		-0.008** (0.003)	0.009* (0.004)
avg_eigenvector		0.017*** (0.003)	0.021*** (0.003)
avg_coherence $\times$ avg_betweenness			0.012*** (0.003)
avg_coherence $\times$ avg_eigenvector			0.005** (0.002)
Num.Obs.	3223	3223	3223
Log.Lik.	4228.41	4443.00	4458.15
AIC	-8448.83	-8871.99	-8898.31
BIC	-8424.51	-8829.45	-8843.61
Sigma	0.181	0.162	0.161
Res.Var	0.025	0.023	0.022
EDF Total	4	7	9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

6 Summary

In summary, this book has no content whatsoever.

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